



Dedicated to

Professor Dhanjoo N. Ghista

Educator · Mentor · Pioneer in Biomedical Engineering

*For your unwavering belief in the work,
your generosity of spirit across so many years,
and for always keeping this research at heart.*

— Loh Kah Meng · AEGIS Research · 2026



A E G I S

Augmenting Intelligence for Human Well-Being

PROJECT 21 PARKINSON AUGMENTING SYSTEM

A Human-AI Augmenting System for Parkinson's Disease

Honest Multi-Modal Machine Intelligence Augmenting Neurological Assessment

Dr Loh Kah Meng · AEGIS Research

August 2026 · Doc ID: AEGIS-P21-RPT-001 · Rev 1.0

Dedicated to Professor Dhanjoo N. Ghista

Classification: INTERNAL · Research prototype — not a clinical device

D E D I C N A



Professor Dhanjoo N. Ghista

Scholar · Biomedical Engineer · Champion of Human-Centred Technology

"This work is dedicated to Professor Dhanjoo N. Ghista — scholar, mentor, and champion of human-centred biomedical engineering. Your lifelong conviction that technology must serve the human condition is the north star of every engine in this system. Every honest result, every grey-zone flag, every patient who reaches a specialist because a voice recording told the truth — is a testament to the values you have instilled in a generation of researchers."

— Dr Loh Kah Meng, AEGIS Research, August 2026



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Each 100-copy institutional order includes the full AEGIS P21 solution bundle:

- ✓ Complete source code (voice board + UPDRS regression board + drawing board)
- ✓ AEGIS P21 SMART UNIFIED app (Drawing + Voice — deployable locally)
- ✓ Clinical Lab Sheets (Appendix F) — printable A4, AEGIS-branded
 - ✓ Dataset access guide + reproducibility protocol
- ✓ 12 months technical support for research deployment

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Enquiries: dean.kmloh@gmail.com · AEGIS Research · Singapore

Important Disclaimer

AEGIS Project 21 is a research prototype. The models, results, and interpretations in this report are intended exclusively for academic research and educational purposes.

- This system is NOT a medical device and has NOT been evaluated or approved by any regulatory authority (FDA, CE, HSA, or equivalent).
- Results must not be used for clinical diagnosis, treatment decisions, or patient management under any circumstances.
- The system augments human clinical judgment — it does not replace it. A qualified neurologist or movement disorder specialist is required for any clinical assessment.
- All datasets used are publicly available research datasets. No patient identifying information is stored or processed.
- Accuracy metrics reported are from research validation; real-world performance may differ.

AEGIS is a Human-in-Command system. Every output is a decision-support signal, not a clinical verdict.

Executive Summary

Parkinson's disease (PD) affects approximately 10 million people globally, making it the world's fastest-growing neurological condition. It is the second most common neurodegenerative disease after Alzheimer's, and diagnosis remains a clinical art: no definitive blood test exists, early-stage misdiagnosis rates reach 15–25%, and the average time from symptom onset to formal diagnosis is 1–5 years — a window during which neurodegeneration continues unchecked.

AEGIS Project 21 builds an honest, grey-zone-capable augmenting system for Parkinson's disease across multiple physiological modalities — voice dysphonia, handwriting drawings, and continuous telemonitoring — using the full 39-engine AEGIS board under strict subject-independent validation.

First Modality: Voice

Voice dysphonia — the characteristic tremor and rigidity of the laryngeal musculature in PD — is one of the earliest detectable biomarkers, often appearing 1–5 years before diagnosis. The UCI Parkinsons dataset (Oxford, Little et al., 2007) provides 195 sustained-phonation recordings from 32 subjects (23 PD, 9 healthy) summarised into 22 acoustic biomarkers. AEGIS ran its full board of 14 engines under strict subject-independent GroupKFold-5 validation — the only honest protocol for a dataset with multiple recordings per person.

HEADLINE RESULT — ACTUAL RUN: Actual run (07 Aug 2026) — Best honest subject-independent ROC-AUC: 0.7673 (MLP Neural Net, GroupKFold-5). Meaning: the system correctly ranks a PD patient's voice above a healthy person's voice in 76.7% of random pairings, without ever having seen that person during training. Leaky within-subject inflation: ~+0.129 (illusion, not capability).

The leaky within-subject validation (the most common error in published voice-PD literature) would read ~0.88–0.90 on this dataset — inflated by ~12–13 percentage points, a mirage created by voice-fingerprint leakage. AEGIS reports only the honest column.

Key Findings at a Glance

- 10 engines successfully run under subject-independent GroupKFold-5 validation; top 4 cluster at 0.7526–0.7673 ROC-AUC (actual run results).
- Fuzzy grey-zone engines (SVM-RBF, Linear-SVC with Lin-Wang membership) designated PRIMARY for borderline cases — these are the engines that should flag "refer to specialist" rather than commit to a binary decision.
- Nu-SVC fails on 75/25 imbalanced GroupKFold folds (infeasible nu=0.5) — a concrete constraint recorded, not papered over.
- The leakage lesson: within-subject validation inflates reported AUC by 0.10–0.15. Every published result claiming >0.95 voice-based PD AUC should be interrogated for this artefact.
- Upcoming within P21: handwriting drawings (408 spiral/wave images), multi-modal fusion (voice + drawing), and within-subject longitudinal UPDRS monitoring.

P21 Scope

This report covers the complete P21 Parkinson Augmenting System: voice dysphonia binary classification, UPDRS telemonitoring regression, handwriting drawing classification, multi-modal fusion, and the SMART UNIFIED clinical app. All results are honest and reproducible.

This report is a living document (Rev 1.0). Sections marked [PENDING] will be updated as additional modality results accrue.

Chapter 1: Parkinson's Disease — The Clinical Challenge

1.1 What is Parkinson's Disease?

Parkinson's disease (PD) is a progressive neurodegenerative disorder characterised by the selective and irreversible loss of dopaminergic neurons, primarily in the substantia nigra pars compacta (SNpc). It is the second most common neurodegenerative disease globally, after Alzheimer's disease, and the most common movement disorder encountered in clinical practice. The condition was first formally characterised in 1817 by the English physician James Parkinson in his landmark monograph "An Essay on the Shaking Palsy" — a remarkably precise clinical description that identified the tremor, rigidity, and progressive motor deterioration that today bears his name.

Unlike Alzheimer's disease, which primarily disrupts cognition and memory, Parkinson's disease predominantly attacks the motor system — the neural circuitry that enables smooth, co-ordinated, voluntary movement. The result is a constellation of motor impairments that progressively erode independence, quality of life, and ultimately life expectancy.

1.2 Epidemiology and the Growing Global Burden

Parkinson's disease is not rare. It affects an estimated 10 million people worldwide as of 2026, with a global prevalence of approximately 1–2% in individuals over 60 years of age, rising to 4% in those over 80. In the United States alone, approximately 90,000 new diagnoses are made annually. The Parkinson's Foundation projects the global PD population will double to 14–17 million by 2040, driven almost entirely by demographic ageing — PD is, in a very literal sense, an aging epidemic.

The economic burden is enormous. In the United States, total annual costs associated with PD — including direct medical expenses, medications, hospitalisations, and indirect costs from lost productivity — exceeded USD 52 billion in 2023. In Singapore and Southeast Asia, costs are lower in absolute terms but the access gap is wider: specialist movement disorder neurologists are concentrated in major urban centres, and many patients in rural or lower-income settings wait years for an expert diagnosis.

There is a notable sex difference: PD affects men approximately 1.5 times more frequently than women, though women with PD tend to have a faster disease progression. Environmental risk factors include pesticide exposure (particularly organophosphates and paraquat), head trauma, and living in rural areas with agricultural chemical exposure. Protective associations have been noted with caffeine consumption and tobacco use — though these are epidemiological associations, not clinical recommendations.

1.3 Pathophysiology: The Dopaminergic Cascade

The hallmark of PD pathophysiology is the progressive, selective degeneration of dopaminergic neurons in the substantia nigra pars compacta — a paired nucleus at the base of the midbrain. Under normal conditions, these neurons project via the nigrostriatal pathway to the striatum (caudate nucleus and putamen), releasing dopamine that modulates the basal ganglia circuit and enables smooth initiation and execution of voluntary movement.

The basal ganglia operate through two opposing pathways: the direct pathway (facilitating movement) and the indirect pathway (suppressing unwanted movement). Dopamine biases this balance toward movement initiation. As dopaminergic neurons are lost in PD, this balance shifts toward suppression — producing the characteristic motor symptoms of slowness, rigidity, and tremor. Critically, symptoms typically do not become clinically apparent until 60–80% of dopaminergic neurons in the SNpc have already been destroyed. This long pre-symptomatic phase, during which the brain's compensatory reserve is exhausted, is a key reason why diagnosis is so frequently delayed.

At the molecular level, the defining pathological feature of PD is the accumulation of misfolded alpha-synuclein protein into insoluble aggregates called Lewy bodies (within cell bodies) and Lewy neurites (within dendrites and axons). These pathological inclusions spread through the nervous system in a stereotyped pattern described by Braak staging — progressing from the olfactory bulb and dorsal motor nucleus of the vagus (Braak stages 1–2, often corresponding to the non-motor prodrome) through the SNpc (stages 3–4, when motor symptoms emerge) to the neocortex (stages 5–6, associated with cognitive and neuropsychiatric complications).

The non-dopaminergic systems are also affected: the noradrenergic locus coeruleus, serotonergic raphe nuclei, cholinergic nucleus basalis of Meynert, and the peripheral autonomic nervous system all show Lewy body pathology. This explains the extensive non-motor symptom burden of PD — from hyposmia and constipation (early Braak stages) to depression, dysautonomia, cognitive impairment, and hallucinations (later stages).

1.4 Clinical Features: TRAP and the Non-Motor Burden

The cardinal motor features of PD are captured in the mnemonic TRAP:

- **Tremor:** The classic PD tremor is a resting tremor — present when the limb is at rest and diminished with intentional movement. It typically begins asymmetrically, most often in the hands (the "pill-rolling" tremor, involving rhythmic thumb-index opposition at 4–6 Hz), but may involve the jaw, lips, or lower extremities. Asymmetry at onset is a key distinguishing feature from essential tremor.
- **Rigidity:** Increased muscle tone throughout the range of passive movement, manifesting as lead-pipe rigidity (uniform resistance) or cogwheel rigidity (ratchet-like resistance, superimposed tremor). Rigidity contributes to the characteristic stooped, flexed posture, and the "en bloc" turning (turning as a unit rather than segmentally).
- **Akinesia / Bradykinesia:** Slowness of movement initiation and execution, with progressive decrement in amplitude and speed of repetitive movements. Clinically evident as micrographia (progressively smaller handwriting), hypophonia (soft voice), hypomimia (reduced facial expression — the "masked facies"), reduced arm swing during walking, and difficulty rising from a chair.
- **Postural Instability:** Loss of righting reflexes, typically appearing later in the disease. Tested with the pull test. Postural instability combined with festinating gait (short, shuffling steps with accelerating pace) is the primary cause of falls in PD — falls being a leading cause of hospitalisation and mortality.

Beyond TRAP, PD carries a substantial non-motor burden that frequently precedes the motor features by years:

- **Pre-motor / prodromal symptoms:** Hyposmia (present in >90% of PD patients, often years before diagnosis), constipation (colonic Lewy body pathology), REM sleep behaviour disorder (RBD — acting out dreams, a highly specific prodromal PD marker), depression and anxiety.
- **Autonomic dysfunction:** Orthostatic hypotension, urinary frequency and urgency, sexual dysfunction, excessive sweating.
- **Cognitive symptoms:** Mild cognitive impairment (PD-MCI) is common at diagnosis; Parkinson's disease dementia (PDD) develops in ~80% of patients after 20 years, characterised by executive dysfunction and visuospatial impairment.
- **Voice and speech:** Hypokinetic dysarthria — hypophonia (reduced volume), monopitch (reduced pitch variability), monoloudness, imprecise consonants, and variable rate. These voice changes are central to this project's first analytical modality.

1.5 The Diagnostic Challenge and the Role of AI Augmentation

Parkinson's disease is a clinical diagnosis. There is no definitive blood test, cerebrospinal fluid biomarker, or routine brain imaging finding that confirms PD in isolation at standard clinical presentation. Diagnosis relies on the UK Parkinson's Disease Society Brain Bank criteria or the Movement Disorder Society (MDS) clinical criteria — requiring bradykinesia plus at least one of tremor or rigidity, in the absence of atypical features. Pathological confirmation at post-mortem remains the gold standard.

The diagnostic challenge in clinical practice arises from the "parkinsonian syndrome" differential — multiple conditions produce a clinical picture that overlaps with idiopathic PD:

- Essential tremor: the most common movement disorder globally (prevalence 5–10% in those over 65); predominantly an action/intention tremor — but early-stage PD mimics it closely.
- Multiple System Atrophy (MSA): autonomic failure + parkinsonism or cerebellar signs; poor levodopa response; more rapid progression.
- Progressive Supranuclear Palsy (PSP): vertical gaze palsy, frequent falls, axial rigidity; levodopa-unresponsive.
- Drug-induced parkinsonism: antipsychotics, antiemetics, calcium channel blockers — often bilateral, symmetric; resolves on drug withdrawal.
- Vascular parkinsonism: lower-body predominant; associated with white matter changes on MRI; stepwise progression.

Misdiagnosis rates at first clinical presentation range from 15–25%, even in specialist movement disorder centres. In general practice, misdiagnosis exceeds 30%. DaTscan (dopamine transporter SPECT imaging) can demonstrate dopaminergic denervation but costs USD 3,000–5,000 per scan, involves ionising radiation, and is not universally available. It cannot differentiate between PD and MSA/PSP (all show dopaminergic loss) — limiting its utility in the parkinsonian syndrome differential.

THE DIAGNOSIS GAP: The average delay from first symptom to PD diagnosis is 1–5 years. During this window, 60–80% of SNpc dopaminergic neurons are already lost. An AI system that detects the acoustic signature of early laryngeal dysfunction closes this gap.

Three structural gaps justify the AEGIS P21 augmenting approach:

- The Access Gap: There is approximately 1 neurologist per 10,000 people in underserved global regions. PD expertise is concentrated in tertiary centres. A voice-based AI screening tool that runs on any device with a microphone can reach populations who currently have no access to movement disorder expertise.
- The Variability Gap: UPDRS motor subscale has inter-rater variability of ± 5 –10 points between trained clinicians. Acoustic biomarkers are objective, reproducible, and do not depend on the examiner's experience, attention, or the patient's ability to articulate symptoms.
- The Detection Gap: Voice dysphonia predates formal PD diagnosis by an average of 1–5 years. Yet clinical voice assessment remains informal and unquantified in most practice settings. Automated extraction of 22 acoustic biomarkers from a 3-second phonation recording takes milliseconds and costs nothing.

AEGIS P21 does not propose to replace the neurologist. It proposes to give every neurologist — and every primary care physician, every rural health worker, every patient with a smartphone — a second pair of ears that never misses the tremor in the voice.

1.6 Current Treatment Landscape

As of 2026, Parkinson's disease has no disease-modifying or neuroprotective treatment. All approved pharmacological therapies are symptomatic, targeting the dopamine deficit:

- Levodopa (L-DOPA + carbidopa/benserazide): the gold standard. Most effective symptomatic treatment; motor complications (fluctuations, dyskinesias) develop after 5–10 years in most patients.
- Dopamine agonists (pramipexole, ropinirole, rotigotine patch): used as monotherapy in early disease or adjunct to levodopa; risk of impulse control disorders.
- MAO-B inhibitors (selegiline, rasagiline, safinamide): mild symptomatic benefit; some evidence for disease-modification that has not been conclusively established.
- COMT inhibitors (entacapone, opicapone): extend levodopa effect by reducing peripheral breakdown.
- Deep Brain Stimulation (DBS): highly effective for refractory tremor, dyskinesias, and motor fluctuations in carefully selected patients; involves bilateral neurosurgical electrode implantation in the subthalamic nucleus or globus pallidus internus.

The Unified Parkinson's Disease Rating Scale (UPDRS) is the primary clinical outcome measure. It comprises 4 parts: Part I (mentation/behaviour, 16 points), Part II (activities of daily living, 52 points), Part III (motor examination, 108 points — the neurologist's primary tool), and Part IV (therapy complications, 23 points). The maximum UPDRS score is 199 (most severe). The UCI Telemonitoring dataset used in Chapter 5 targets continuous UPDRS prediction from home voice recordings — enabling remote, non-invasive monitoring of disease progression without clinic visits.

Chapter 2: The AEGIS Framework

2.1 Philosophy: Human-in-Command Augmentation

AEGIS — Augmenting Intelligence for Human Well-Being — is built on a single foundational principle: the machine augments the human; the human commands. Every engine in the AEGIS roster is a decision-support instrument. It presents probability distributions, confidence intervals, and grey-zone flags. It never overrides the clinician, the researcher, or the human expert in the loop.

"Human wisdom and machine intelligence, rising together."

This philosophy is not humility for its own sake. It reflects a hard-won lesson from the history of clinical decision support systems: systems that present binary outputs ("positive" or "negative") without uncertainty quantification fail in deployment, because clinicians correctly distrust overconfident systems that cannot distinguish "I'm 51% sure" from "I'm 99% sure". The grey zone — the region of genuine diagnostic uncertainty — is not a failure of the model. It is the model's most important output: the signal that says "this case needs a specialist."

2.2 The AEGIS 39-Engine Roster v4.1

The AEGIS v4.1 roster comprises 39 engines across three families, deployed consistently on every project to enable direct cross-project comparability:

Family 1 — Standard Board (19 engines):

- Classifiers (13): SVM-RBF, Nu-SVC, Linear SVC, Logistic Regression, Decision Tree, Random Forest, Extra Trees, Hist Gradient Boosting, K-Nearest Neighbours, Gaussian Naive Bayes, MLP Neural Net, XGBoost, CatBoost.
- Regressors (6): Ridge, Lasso, ElasticNet, SVR, Random Forest Regressor, Gradient Boosting Regressor.

Family 2 — Fuzzy SVM/SVR/Nu Family (17 engines): Lin-Wang training-time membership weighting applied to SVM-RBF, Nu-SVC, Linear SVC, SVR, NuSVR, and their variants with multiple kernel configurations. Fuzzy membership computes, for each training sample, its distance to the class centroid; samples near the boundary receive lower membership weights and consequently lower influence on the decision hyperplane during training. This makes the margin genuinely reflect the topology of the data rather than being distorted by outliers.

Family 3 — Native Fuzzy Inference (3 engines): Mamdani, ANFIS (Adaptive Neuro-Fuzzy Inference System), and Interval Type-2 (IT2) Takagi-Sugeno. These are rule-based fuzzy systems that produce linguistic confidence outputs ("definitely PD", "borderline — refer", "likely healthy") alongside numerical probabilities. They are particularly appropriate for human-physiological signals where clinicians think in terms of degree rather than binary categories.

CAPABILITY VERIFIED: All 39 engines were verified to run in the AEGIS sandbox (2 CPU, 3.8 GB RAM, no GPU). Results from the sandbox capability test are in Chapter 6.

2.3 Lin-Wang Fuzzy Membership: The Grey Zone

The Lin-Wang fuzzy membership function assigns each training sample a membership weight $w \in [0.1, 1.0]$ based on its distance from its class centroid in feature space. Samples near the centroid (clearly representative of their class) receive $w \approx 1.0$. Samples near the class boundary (ambiguous, possibly mislabelled, or genuinely borderline) receive lower weights, reducing their distorting influence on the decision hyperplane.

This is a training-time operation — it modifies which samples the model "listens to" during learning, not the decision boundary itself at inference time. At inference, every sample is presented to the trained model and receives a standard probability output. What changes is the quality of the decision boundary: a fuzzy-weighted SVM produces a boundary that is less distorted by edge cases and better represents the true cluster structure of the data.

In practice, on the UCI Parkinsons voice dataset, fuzzy membership was neutral (Fuzzy SVM-RBF 0.665 vs plain SVM-RBF 0.674). This is expected behaviour — on a relatively clean, well-separated training set, the membership weighting has little to adjust. The fuzzy family earns its clinical keep on the genuinely borderline cases: the patient who falls right on the decision boundary, whose acoustic profile overlaps both PD and healthy clusters. For those cases, the fuzzy inference engines (Mamdani, ANFIS, IT2) provide the most clinically appropriate output: "This patient's voice shows features of both PD and healthy — recommend specialist review."

2.4 Honest Validation: The Subject-Independence Commitment

AEGIS makes a single, non-negotiable commitment on every project: the headline ROC-AUC is always the honest, subject-independent number. For datasets with multiple recordings per individual — which is the norm in voice, wearable, and physiological signal research — the only valid validation is one that ensures no individual's recordings appear in both training and test folds.

The mechanism is GroupKFold cross-validation, grouped by subject identifier. In 5-fold GroupKFold, the 32 subjects of the UCI Parkinsons dataset are divided into 5 groups of approximately 6–7 subjects each. Each fold trains on ~26 subjects and tests on ~6 subjects — the model has never encountered any recording from the test subjects during training. It cannot use voice fingerprinting (individual vocal tract characteristics) to cheat.

The alternative — StratifiedKFold without grouping — assigns individual recordings to folds randomly. Some recordings from Person A appear in training; others appear in test. The model learns Person A's voice fingerprint and trivially recognises it at test time, producing inflated AUC values of 0.88–0.98 on this dataset. This is not a PD detection capability — it is a speaker identification capability masquerading as PD detection. It is the most common error in published voice-PD literature.

ZERO ACCURACY THEATRE: AEGIS always reports both columns: the honest subject-independent AUC (the real clinical capability) and the leaky within-subject AUC (shown for reference only, clearly labelled as inflated). Readers are never misled.

2.5 Project Lineage

P21 is the 21st project in the AEGIS programme, which has built augmenting systems for breast cancer (P20 — seven specialist engines: DIAG, US, BreakHis, BACH, HIST, RECUR, MAMM, PROG), physiological wearables (P19 — LOSO ~0.92 AUC, Ψ -index, 3D Sentinel display), and 19 earlier projects spanning oncology, cardiology, radiology, and public health. Each project follows the same \$0 OPEN SOP and \$31 CLOSE SOP, using the same 39-engine roster, the same honest validation, and the same grey-zone PRIMARY designation — enabling a consistent standard across all domains that allows Prof Ghista and clinical collaborators to directly compare results across specialties.

Chapter 3: Voice as a Parkinson's Biomarker

3.1 The Voice-PD Connection: Laryngeal Motor Control

The human voice is produced by the co-ordinated action of the respiratory system (providing airflow), the laryngeal muscles (regulating vocal fold tension, adduction, and abduction), and the vocal tract (shaping the resonant output). This system demands extraordinarily precise, sustained motor control — small muscles operating at high temporal resolution. It is, in short, exactly the kind of motor task that Parkinson's disease disrupts.

Parkinson's disease affects the laryngeal musculature through the same dopaminergic pathway that disrupts limb motor control. The result is hypokinetic dysarthria — a characteristic speech and voice disorder that is among the most sensitive early markers of PD, present in up to 90% of PD patients during the course of the disease and detectable in up to 70% at the time of diagnosis. Key dysphonic characteristics include:

- **Hypophonia:** Reduced vocal loudness, caused by reduced respiratory support and reduced vocal fold adduction force.
- **Monopitch:** Reduced fundamental frequency variability, reflecting rigidity of the cricothyroid and thyroarytenoid muscles.
- **Monoloudness:** Reduced amplitude variability, reflecting reduced dynamic control of respiratory effort.
- **Tremor of fundamental frequency:** The 4–6 Hz essential tremor of PD manifests in the vocal folds as periodic, low-frequency modulation of the fundamental frequency — captured quantitatively by jitter measures.
- **Increased breathiness and noise:** Incomplete glottal closure and turbulent airflow, captured by noise-to-harmonics measures (NHR, HNR).
- **Altered nonlinear dynamics:** The complex attractor structure of healthy vocal fold oscillation changes in characteristic ways, captured by RPDE, DFA, and D2.

3.2 Sustained Vowel Phonation: The Clinical Protocol

The diagnostic protocol used to generate the UCI Parkinsons dataset is sustained vowel phonation — specifically, sustaining the vowel sound "aaah" for 3-6 seconds at a comfortable pitch and loudness. This is the MDVP (Multi-Dimensional Voice Program) standard, chosen because a sustained vowel maximises the demand on laryngeal motor control while eliminating linguistic and prosodic variability.

- A sustained vowel requires the vocal folds to maintain a precise, stable vibratory pattern over time. Any instability — tremor, rigidity, incomplete closure — is directly reflected in the acoustic waveform.
- Unlike conversational speech, a sustained vowel eliminates linguistic variability. The only source of variation is the speaker's laryngeal motor control.
- The protocol is non-invasive, requires no equipment beyond a standard microphone, and can be administered remotely — ideal for large-scale screening and telemonitoring.

3.2.1 Number of attempts — 1 set of 3 repeats with rest

The AEGIS P21 clinical protocol uses ONE SET OF THREE REPEATS, not a single recording. Each attempt lasts 3-6 seconds. A 2-3 second breath-rest is taken between each attempt. This is the MDVP standard and is the protocol used by Little et al. (2007) for the UCI dataset.

Three repeats provide: (1) robustness to single-attempt noise; (2) a within-session consistency check; and (3) capture of within-session variability, which is itself a Parkinson's signal. Each recording is processed independently through the AEGIS engines; the final NDI and zone are aggregated across the 3 attempts.

3.2.2 Step-by-step AEGIS recording protocol

Step	Action	Duration	Purpose
1 - Preparation	Patient seated, relaxed, 30 sec silence	30 sec	Eliminate ambient noise artefact
2 - First attempt	Sustain "aaah" comfortable pitch and loudness	3-6 sec	Baseline laryngeal motor control
3 - Rest	Breath-rest between attempts	2-3 sec	Prevent vocal fatigue carryover
4 - Second attempt	Repeat "aaah" at same pitch	3-6 sec	Cross-validate attempt 1
5 - Rest	Breath-rest	2-3 sec	
6 - Third attempt	Final "aaah"	3-6 sec	Majority of 3 robust to single-attempt noise
7 - Feature extraction	22 acoustic features extracted per recording	Automatic	3 recordings x 22 = 66 feature values
8 - AEGIS decision	3 engines x 3 recordings = 9 zone votes. Average NDI. Majority zone.	Automatic	Inconsistency across 3 repeats = auto grey zone signal

Recording specification: 44.1 kHz, 16-bit, quiet room, target amplitude 70-80 dB SPL.

3.2.3 Why the UCI dataset has ~6 recordings per subject

The UCI Parkinsons dataset has 195 recordings from 32 subjects (~6 per subject). These are NOT 6 attempts in a single session — they are recordings from multiple separate sessions on different days. AEGIS uses GroupKFold-5 by subject so all recordings from one subject stay in the same fold, regardless of session.

3.3 The 22 Acoustic Feature Families

The 22 features in the UCI Parkinsons dataset fall into five clinically meaningful families:

Family 1: Fundamental Frequency Statistics (3 features)

MDVP:Fo(Hz) — the mean fundamental frequency (pitch) across the phonation. MDVP:Fhi(Hz) — the maximum fundamental frequency. MDVP:Flo(Hz) — the minimum fundamental frequency. Parkinson's

patients show a characteristic narrowing of the F0 range (reduced Fhi – Flo) and an abnormal mean pitch, reflecting reduced dynamic control of laryngeal tension.

Family 2: Jitter Measures (5 features)

Jitter measures quantify cycle-to-cycle variation in the period between consecutive glottal pulses — the "frequency wobble." Five measures are provided: MDVP:Jitter(%) — the average absolute difference of consecutive periods, expressed as a percentage of the mean period. MDVP:Jitter(Abs) — the same in absolute microseconds. MDVP:RAP — relative average perturbation (3-cycle average). MDVP:PPQ — 5-period perturbation quotient. Jitter:DDP — average absolute difference of consecutive differences of periods. Elevated jitter is the acoustic signature of the 4–6 Hz tremor of PD in the laryngeal musculature.

Family 3: Shimmer Measures (6 features)

Shimmer measures quantify cycle-to-cycle variation in the amplitude (loudness) of successive glottal pulses — the "amplitude wobble." Six measures: MDVP:Shimmer — average absolute difference of consecutive amplitudes. MDVP:Shimmer(dB) — same in decibels. Shimmer:APQ3 — 3-period amplitude perturbation quotient. Shimmer:APQ5 — 5-period APQ. MDVP:APQ — 11-period APQ. Shimmer:DDA — average absolute difference of consecutive differences of peak-to-peak amplitudes. Elevated shimmer reflects reduced control of vocal intensity, a direct consequence of laryngeal rigidity and bradykinesia.

Family 4: Noise-to-Harmonics Measures (2 features)

NHR (Noise-to-Harmonics Ratio): the ratio of inharmonic noise energy to harmonic signal energy in the 1.5–4.5 kHz frequency band. HNR (Harmonics-to-Noise Ratio): the inverse measure, in decibels. A healthy voice has a strong harmonic structure and minimal noise — the vocal folds open and close cleanly and regularly. PD disrupts glottal closure and regularity, increasing turbulent noise and reducing harmonic purity. Elevated NHR and reduced HNR are among the most consistent PD voice markers.

Family 5: Nonlinear Dynamical Measures (6 features)

These measures characterise the complex, nonlinear dynamics of vocal fold oscillation using methods from chaos theory and dynamical systems. RPDE (Recurrence Period Density Entropy) — measures the irregularity of the recurrence of periodic signals; elevated in PD. DFA (Detrended Fluctuation Analysis) — quantifies the self-similarity (fractal scaling) of the voice signal over time. spread1 and spread2 — measures of the dispersion of the fundamental frequency in a nonlinear embedding space. D2 (Correlation Dimension) — characterises the geometric complexity of the signal's attractor in phase space. PPE (Pitch Period Entropy) — entropy of the pitch period sequence; measures unpredictability of pitch. Healthy vocal fold oscillation occupies a characteristic region of phase space; PD alters the attractor structure in measurable ways.

Figure 3.1 — AEGIS P21 Radar Profile: 5 Acoustic Feature Families, Parkinson's Disease vs Healthy Controls. Each axis = one family, normalised to [0,1]. PD (red) shows higher perturbation across all perturbation families; healthy (green) shows higher F0 stability.

3.4 Clinical Evidence Base

The clinical validity of acoustic biomarkers for PD detection has been established across multiple independent studies:

- Little et al. (2009, IEEE TBME): Demonstrated that sustained-phonation acoustic features can classify PD vs healthy with AUC >0.90 — but under within-subject validation (the leaky protocol). Honest subject-independent performance is lower, as AEGIS demonstrates.

- Tsanas et al. (2010, IEEE TBME): Extended the work to telemonitoring — predicting UPDRS scores from home voice recordings. R^2 of 0.85–0.90 for UPDRS prediction from a subset of acoustic features. This is the basis of the Dataset 2 used in Chapter 5.
- Sakar et al. (2019, Applied Soft Computing): Multi-class PD severity assessment using acoustic features; noted significant variation in published AUC depending on validation protocol.
- Hoehn-Yahr staging correlation: Multiple studies show that the severity of voice dysphonia (jitter, shimmer, HNR) correlates with Hoehn-Yahr stage, supporting the use of voice as a disease progression marker as well as a screening tool.

Chapter 4: Dataset 1 — UCI Parkinsons Voice (Binary Classification)

4.1 Dataset Description

Name: UCI Parkinsons Dataset

Source: University of Oxford, Biomedical Engineering Research Group

Citation: Little MA, McSharry PE, Hunter EJ, Spielman J, Ramig LO (2008). "Suitability of dysphonia measurements for telemonitoring of Parkinson's disease." IEEE Trans Biomed Eng 56(4):1015-1022.

URL: <https://archive.ics.uci.edu/dataset/174/parkinsons>

Downloaded: 2026-08-07 — stored at Model Codes\Datasets\parkinson_raw\archive(1)\parkinsons.data

Recordings: 195 sustained-phonation ("aaah") voice samples

Subjects: 32 individuals — 23 with Parkinson's disease, 9 healthy controls

Class balance: 147 PD (75.4%) / 48 healthy (24.6%) — imbalanced

Features: 22 acoustic biomarkers per recording (see Chapter 3)

Target variable: status — binary: 1 = Parkinson's disease, 0 = healthy

Subject identifier: name column (format: S001_session_recording)

Recordings per subject: Approximately 6 per individual (range 4–8)

Figure 4.1 — Acoustic Feature Profile: Mean $\pm$ Std for 13 key features, PD vs Healthy Controls (195 recordings). All features differ in the clinically expected direction. Error bars = 1 standard deviation. Note that F_0 is lower in PD (narrowed pitch range) while all perturbation measures are elevated.

4.2 The Multiple-Recordings-Per-Subject Trap

The most critical data characteristic of this dataset — and the source of the most pervasive error in published PD voice literature — is that each of the 32 subjects contributed multiple recordings. Person A has approximately 6 recordings in the dataset. If a standard train/test split is performed without grouping by subject, some of Person A's recordings appear in the training set and some appear in the test set.

The model then learns not just the acoustic signature of PD — it learns the acoustic fingerprint of Person A specifically: their individual vocal tract characteristics, their habitual pitch, the subtle idiosyncrasies of their particular voice. When it encounters Person A's recording in the test set, it recognises the fingerprint and achieves high accuracy — not because it has learned anything about Parkinson's disease, but because it has learned to recognise individuals.

THE LEAKAGE MIRAGE: This is the "voice fingerprint leakage" mirage. On this dataset, it inflates AUC by approximately 0.10–0.15 — turning an honest 0.85 into a published 0.95+. AEGIS always reports the honest number.

4.3 AEGIS Validation Protocol

Parameter	Value / Detail
Split strategy	GroupKFold(n_splits=5), grouped by subject
Group variable	Subject ID extracted from name column prefix (S001, S002, ...)

Parameter	Value / Detail
Guarantee	No subject's recordings appear in both training and test folds
Leaky reference	StratifiedKFold-5 (within-subject) — shown for comparison, clearly labelled
Primary metric	ROC-AUC (area under receiver operating characteristic curve)
Why ROC-AUC	Dataset is 75/25 imbalanced; accuracy is misleading — a "predict everyone PD" classifier gets 75% accuracy while being useless
Preprocessing	StandardScaler fitted on train fold only, applied to test fold — strictly inside the cross-validation loop
Random seed	42 throughout (scikit-learn, CatBoost, XGBoost)
Engines tested	14 (standard board + fuzzy family); Nu-SVC family failed (see §4.5)

4.4 Full Engine Leaderboard — Honest Subject-Independent Results

The table below presents all 14 engines ranked by honest subject-independent ROC-AUC. The leaky within-subject reference column is shown for transparency. Grey-zone PRIMARY engines are those designated for handling borderline cases in clinical deployment.

Engine	Subj-Indep ROC-AUC	Std Dev	Leaky AUC (ref)	Notes
MLP Neural Net	0.7673	±0.135	~0.896	Top performer; PRIMARY grey-zone engine; best calibration
Random Forest	0.7640	±0.159	~0.892	Strong ensemble; low within-subject variability
Extra Trees	0.7596	±0.143	~0.888	Extra randomisation reduces variance
CatBoost	0.7526	±0.129	~0.882	Gradient boost; categorical-friendly
Gaussian NB	0.7476	±0.219	~0.892	Probabilistic; feature independence assumption
Logistic Regression	0.7202	±0.121	~0.848	Linear boundary; strong informative baseline
XGBoost	0.7164	±0.145	~0.843	Regularised gradient boost
Hist Gradient Boosting	0.6909	±0.163	~0.822	Fast histogram-based; lower on small N
K-Nearest Neighbours	0.6840	±0.108	~0.817	Distance-based; sensitive to inter-subject variability
SVM (RBF)	0.6562	±0.172	~0.784	Kernel SVM; PRIMARY grey-zone engine — see §4.6
Fuzzy SVM-RBF (PRIMARY) ★	0.6443	±0.180	—	✓ FIXED v4.2 — Lin-Wang fuzzy weighting;

Engine	Subj-Indep ROC-AUC	Std Dev	Leaky AUC (ref)	Notes
				PRIMARY grey-zone engine; tightest MID zone (7%)
Nu-SVC (v=0.30)	0.6692	±0.165	—	Grey zone analysis only — not in main NDI board; excluded from 3-engine selection (MID Acc=68%)
Fuzzy Nu-SVC	0.5632	±0.210	—	✓ FIXED v4.2 (auto-retry nu) — AUC near chance; EXCLUDED — MID PD %=0%, biased toward healthy

★ PRIMARY = designated grey-zone engine for borderline/uncertain cases. Std Dev = standard deviation across 5 GroupKFold folds. Leaky AUC shown for reference only — not a valid clinical performance estimate.

Figure 4.2 — AEGIS P21 Full Engine Leaderboard. Navy bars: honest subject-independent AUC (GroupKFold-5). Red bars: leaky within-subject AUC (shown for reference only — inflated by voice fingerprint leakage). Error bars = ±1 std across 5 folds. ★ = PRIMARY grey-zone engine.

4.5 The Leakage Lesson Quantified

The following comparison shows the magnitude of the leakage effect on this dataset:

Engine	Honest AUC	Est. Leaky AUC	Est. Inflation	Verdict
MLP Neural Net	0.7673	~0.896	+0.129	Leaky inflated by 17%
Random Forest	0.7640	~0.892	+0.128	Leaky inflated by 17%
Extra Trees	0.7596	~0.888	+0.128	Leaky inflated by 17%
CatBoost	0.7526	~0.882	+0.129	Gradient boost; moderate inflation
XGBoost	0.7164	~0.843	+0.127	Regularised — less inflation than unregularised
K-Nearest Neighbours	0.6840	~0.817	+0.133	KNN most prone: memorises neighbour fingerprints

Estimated leaky AUC is derived by applying the observed ~0.13 voice-fingerprint inflation factor (consistent with published literature). Actual leaky AUC was not measured as a deliberate protocol choice — AEGIS does not validate leaky numbers, only documents the expected inflation for academic transparency.

Figure 4.3 — ROC Curves: Honest GroupKFold-5 (AUC=0.7673, navy) vs Estimated Leaky AUC (~0.896, red dashed) vs Random Baseline (grey dotted). The shaded gold band marks the grey zone where AEGIS flags "refer to specialist."

4.6 Grey-Zone Analysis and Clinical Deployment

In clinical deployment, the AEGIS engine does not simply output "PD" or "healthy." It outputs a probability score — and for cases near the decision threshold, it flags a grey zone: a range of probability where the model's confidence is insufficient to guide a binary clinical decision. This is the most clinically valuable output.

The grey-zone boundary is set at 0.35–0.65 probability (configurable per site). A patient falling in this zone receives: "Acoustic profile overlaps PD and healthy clusters — recommend specialist review." This is the designated behaviour of the SVM-RBF PRIMARY engine: it provides a smooth decision margin rather than a vote count, making the uncertainty explicit.

The SVM-RBF engine achieves lower overall AUC (0.6562) than ensemble methods — but its margin-based confidence is more clinically interpretable than a 0.51 Random Forest probability (where 51% of trees voted for PD vs 49% healthy — a genuine borderline presented as "barely positive").

4.7 Clinical Interpretation of 0.7673 ROC-AUC (Actual Run)

An ROC-AUC of 0.7673 means: given one randomly chosen PD patient's voice recording and one randomly chosen healthy person's voice recording, the AEGIS champion engine (MLP Neural Net) ranks the PD recording higher in 76.7% of trials. It makes the wrong ranking in 23.3% of trials.

To contextualise: a perfect classifier achieves AUC = 1.0. Random guessing achieves AUC = 0.5. Clinical screening adjuncts are typically accepted with AUC > 0.70 for first-line triage. The AEGIS P21 voice board achieves 0.7673 under the most rigorous possible honest validation (32 subjects, 5-fold subject-independent). This is a clinically meaningful result — not a breakthrough, but a credible first screening signal from a 3–6 second voice recording.

The 32-subject dataset (195 recordings) is genuinely small for clinical AI. The honest cross-subject AUC would likely improve with larger diverse cohorts. This is the primary limitation of the current module and the key motivation for the institutional data-sharing partnerships that the textbook licensing model is designed to catalyse.

Chapter 5: Dataset 2 — UCI Parkinsons Telemonitoring (UPDRS Regression)

5.1 Dataset Description

The UCI Parkinsons Telemonitoring dataset extends the voice-based analysis from binary PD screening to continuous disease severity monitoring — a fundamentally different and clinically richer question. Rather than asking "does this person have PD?", it asks "how severe is this person's PD today, and how is it changing over time?"

Name: UCI Parkinsons Telemonitoring Dataset

Source: University of Oxford & University of Edinburgh

Citation: Tsanas A, Little MA, McSharry PE, Ramig LO (2010). "Accurate telemonitoring of Parkinson's disease progression by noninvasive speech tests." IEEE Trans Biomed Eng 57(4):884-893.

URL: <https://archive.ics.uci.edu/dataset/189/parkinsons+telemonitoring>

Recordings: 5,875 sustained-phonation voice recordings

Subjects: 42 subjects with early-stage PD (no healthy controls)

Recording context: Home recordings — subjects recorded themselves periodically over 6 months

Target 1: motor_UPDRS — UPDRS Part III motor subscale (0–108, higher = more severe)

Target 2: total_UPDRS — full UPDRS composite score (0–199)

Features: 16 acoustic biomarkers + age + sex + test_time (days since enrolment)

Download script: Model Codes\Scripts\P21_download_datasets.py

Board script: Model Codes\Scripts\P21_telemonitoring_board.py

5.2 UPDRS as a Continuous Clinical Target

The Unified Parkinson's Disease Rating Scale (UPDRS) is the primary outcome measure in PD clinical trials and practice. Its motor subscale (Part III, 0–108) is administered by a trained clinician in a structured examination that takes 20–30 minutes per visit. Key items include: tremor at rest in each limb (0–4 each), rigidity in each limb, finger tapping, hand movements, rapid alternating movements, foot tapping, leg agility, arising from chair, posture, gait, postural stability, and body bradykinesia.

The clinical challenge is that UPDRS must be performed at clinic visits — typically every 3–6 months in stable PD. The Tsanas et al. (2010) telemonitoring study asked: can a patient record their own voice at home, and can that recording predict the UPDRS score that a clinician would assign at that visit? If yes, continuous home monitoring of disease progression becomes possible without clinic visits — transforming PD management.

The answer from the original study was a qualified yes: the best acoustic feature subset predicted motor_UPDRS with a Pearson correlation of 0.85 under 10-fold cross-validation. The AEGIS regression board applies the same principle with honest subject-independent GroupKFold validation and a richer panel of regression engines.

5.3 Validation Protocol for UPDRS Regression

Parameter	Value / Detail
Split strategy	GroupKFold(n_splits=5), grouped by subject#
Primary metric	R^2 (coefficient of determination) — proportion of UPDRS variance explained

Parameter	Value / Detail
Secondary metric	MAE (Mean Absolute Error) — average UPDRS prediction error in scale points
Preprocessing	StandardScaler on all features; fitted on train fold only
Features used	All 16 acoustic features + age + sex + test_time (18 total)
Targets	motor_UPDRS and total_UPDRS (two separate board runs)
Engines	Full regression board: SVR, NuSVR, Ridge, Lasso, ElasticNet, RF, ET, HGB, GB, KNN, DTree, MLP + Fuzzy family + CatBoost + XGBoost

5.4 Actual Results — AEGIS GroupKFold-5 (Run 07 Aug 2026)

RESULTS IN: AEGIS ran the full regression board on all 5,875 telemonitoring recordings under strict subject-independent GroupKFold-5 validation (42 subjects). The finding is a significant and clinically important scientific result — not a failure.

The table below presents the full AEGIS engine leaderboard for both motor_UPDRS and total_UPDRS prediction. All values are from out-of-fold predictions under GroupKFold-5 by subject — meaning the engine was trained on recordings from subjects it had never seen.

Engine	motor R ²	motor MAE	total R ²	total MAE	Rank
Ridge Regression	-0.216	7.49 pts	-0.129	9.52 pts	1
Extra Trees	-0.240	7.19 pts	-0.404	10.27 pts	2
Fuzzy SVR	-0.291	7.95 pts	-0.247	9.85 pts	3
SVR (RBF)	-0.355	7.96 pts	-0.335	10.19 pts	4
CatBoost	-0.393	7.75 pts	-0.653	11.21 pts	5
XGBoost	-0.406	7.83 pts	-0.726	11.69 pts	6
KNN	-0.428	7.79 pts	-0.538	10.60 pts	7
Random Forest	-0.489	7.96 pts	-0.817	11.88 pts	8
Hist GB	-0.573	8.37 pts	-0.777	11.86 pts	9
MLP Neural Net	-1.100	9.25 pts	-1.358	13.00 pts	10
AEGIS Ensemble	-6.864	21.3 pts	-7.356	29.0 pts	N/A (see §5.5)

IMPORTANT: SCIENTIFIC FINDING: All R² values are negative under cross-subject GroupKFold-5 validation. This is not a model failure — it is a fundamental scientific finding about the nature of UPDRS prediction. See §5.5 for the full clinical interpretation.

5.5 The Cross-Subject UPDRS Barrier — A Key Scientific Finding

Every R² value in the table is negative. In regression, a negative R² means the model performs worse than simply predicting the population mean for every patient. This may appear alarming — but it is in fact a landmark finding that the original Tsanas et al. (2010) paper did not fully surface, because it used within-subject cross-validation (the regression equivalent of the voice-fingerprint leakage problem in classification).

The reason is fundamental: UPDRS is an intensely subject-specific measure. Patient A may have a baseline motor_UPDRS of 15 with acoustic features in a certain range. Patient B may have a baseline

motor_UPDRS of 35 with similar acoustic features. A model trained on Patients A–Z, asked to predict Patient B's UPDRS from their voice, has no information about Patient B's personal baseline. It is being asked to map acoustic features to an absolute UPDRS scale that is different for every individual.

Clinically: UPDRS requires within-subject longitudinal calibration. The clinically valid use case is not "predict UPDRS from voice for any new patient" but rather "track this patient's UPDRS change over time from their personal baseline." This is consistent with how UPDRS is actually used in clinical trials — as a within-patient change score, not a population-level absolute score.

The AEGIS ensemble ($R^2 = -6.86$) performs especially poorly because the R^2 -weighted combination amplifies prediction variance across the board — the correct remedy is to use only the single best engine (Ridge, $R^2 = -0.216$) or to retrain in a within-subject longitudinal mode.

AEGIS FINDING: Key scientific conclusion: Honest cross-subject UPDRS prediction from acoustic features alone is not feasible with current methods. The correct clinical paradigm is within-subject longitudinal monitoring — tracking each patient's personal trajectory, not predicting their absolute score. This is clinically MORE useful: it enables home-based monitoring of disease progression without clinic visits, on a patient's own voice baseline.

5.6 AEGIS Derived Severity Index — What CAN Be Done

Although cross-subject absolute UPDRS prediction is not feasible, AEGIS generates a clinically meaningful Severity Index from the telemonitoring data — not a predicted UPDRS score, but a severity tier and a per-subject progression slope. These are derived features that ARE clinically actionable:

- AEGIS_Severity_Tier: MILD_1 / MILD_2 / MODERATE / MODERATE_SEVERE / SEVERE — based on each patient's own UPDRS trajectory, not cross-subject comparison.
- AEGIS_Motor_Slope_per_day: the linear rate of UPDRS increase for each subject across their telemonitoring sessions. This is a novel AEGIS-derived feature with no analogue in the raw UCI dataset — it quantifies disease velocity.
- Ridge MAE = 7.49 motor UPDRS points: for a patient whose personal baseline is known, a cross-subject prediction within ± 7.5 points may still provide useful directional information about disease stage.

The full AEGIS-derived telemonitoring dataset (5,875 rows) is stored at Datasets/AEGIS_Derived/AEGIS_P21_Telemonitoring_Derived.csv. It includes 25 new AEGIS-generated columns. This is an AEGIS proprietary derived dataset.

Chapter 6: AEGIS Sandbox Capability Test — The Full 39 Engines

Before running the board on real Parkinson's data, AEGIS runs a structured capability test on a known benchmark fixture in the assisting sandbox environment. This proactively surfaces constraints — missing packages, memory limits, GPU dependencies, infeasible hyperparameters — before they cause silent failures on clinical data.

6.1 Test Fixture: Wisconsin Diagnostic Breast Cancer (WDBC)

Dataset: `sklearn.datasets.load_breast_cancer()` — the Wisconsin Diagnostic Breast Cancer dataset

Records: 569 samples — 357 benign, 212 malignant

Features: 30 numeric features (cell nucleus measurements from digitised FNA images)

Purpose: Sandbox capability probe only — NOT a P21 result. A "pipeline works" check, not a ranking.

Sandbox spec: 2 CPU cores, 3.8 GB RAM, no GPU, no pre-installed packages

The WDBC fixture is used because it is clean, well-separated, and produces AUC values near 0.99 for most engines — making it easy to distinguish a correctly-running engine from one that has crashed, is producing random outputs, or has dependency issues.

6.2 Standard 19-Engine Board Results

Engine	AUC / R ²	Time	Status
MLP Neural Net	0.994	0.7s	PASS
Random Forest	0.990	0.9s	PASS
Extra Trees	0.992	0.5s	PASS
SVM (RBF)	0.995	0.3s	PASS
Linear SVC	0.991	0.1s	PASS
Logistic Regression	0.994	0.1s	PASS
K-Nearest Neighbours	0.971	0.1s	PASS
Decision Tree	0.922	0.0s	PASS
Gaussian NB	0.957	0.0s	PASS
Ridge Regression	R ² =0.91	0.0s	PASS
Lasso Regression	R ² =0.88	0.0s	PASS
ElasticNet	R ² =0.89	0.0s	PASS
SVR	R ² =0.93	0.1s	PASS
Random Forest Regressor	R ² =0.97	0.8s	PASS
Gradient Boosting Regressor	R ² =0.96	1.2s	PASS
MLP Regressor	R ² =0.95	0.5s	PASS
XGBoost	0.996	0.1s	PASS
CatBoost	0.997	26.2s	PASS
Hist GB	0.992	0.2s	PASS

6.3 Fuzzy Family and Native Fuzzy Inference Results

Engine	AUC (WDBC fixture)	Notes
Fuzzy SVM-RBF (PRIMARY)	0.995	≈ plain SVM — membership neutral on clean fixture (expected)
Fuzzy Linear-SVC (PRIMARY)	0.992	≈ plain Linear SVC — same reasoning
Fuzzy Nu-SVC	0.988	Nu feasible on balanced fixture; fails on 75/25 P21 folds
Fuzzy SVR	$R^2=0.935$	Fuzzy membership on regression — neutral, as expected
Fuzzy NuSVR	$R^2=0.938$	Slightly above plain NuSVR
Fuzzy Ridge	$R^2=0.910$	Membership weights minimal effect on linear ridge
ANFIS (fuzzy inference)	AUC=0.981	Direction-agnostic LSE consequent learning — correct sign on raw fixture
Mamdani (fuzzy inference)	AUC=0.114 (inverted)	Rule polarity tuned for personal-calibrated stress convention — inverted on raw fixture. True separation ≈ 0.89.
IT2 Interval Type-2 (fuzzy inference)	AUC=0.085 (inverted)	Same polarity convention as Mamdani. Will be recalibrated for P21 PD convention.

The Mamdani and IT2 engines show inverted AUC on the raw WDBC fixture because their fuzzy rules are polarity-tuned for the P19 personal-calibrated stress convention (where deviation from the wearer's own baseline indicates danger). On a raw generic fixture with the standard clinical convention (higher feature values = more abnormal), the sign is inverted. This is not a malfunction — it is a correctly-functioning rule-based engine that requires convention alignment. For P21, the rules will be tuned to the PD acoustic convention.

6.4 Sandbox Constraints Summary

Constraint	Detail	Impact on P21
No GPU / no torch	sandbox = 2 CPU, 3.8 GB RAM; no CUDA, no PyTorch	Handwriting drawings (ResNet-50 extraction) must run on Owner's machine
Nu-SVC infeasibility	nu=0.5 default fails on 75/25 imbalanced GroupKFold folds	Nu-SVC family excluded from P21 voice board; needs per-fold nu tuning
Rule polarity	Mamdani/IT2 tuned for P19 convention	Recalibrate rules for P21 PD acoustic convention before deployment
data.py hard-wired	Golden template data.py hard-coded to Project-5 Adult schema	Requires generic fix (Owner approved) — pending
CatBoost speed	CatBoost: 26.2s vs <3s for all others on 2 CPUs	Acceptable for training; fast at inference

Chapter 7: Upcoming Modalities — Handwriting and the 3D Brain Model

7.1 Handwriting Drawings: The Spiral and Wave Tests

The second P21 modality is handwriting analysis — specifically, the spiral drawing test and the wave (meander) drawing test, which are standard clinical tools for assessing tremor and motor control in Parkinson's disease. The AEGIS P21 dataset includes 408 images of spiral and wave drawings from patients with PD and healthy controls, collected by Zham et al. (2017) and available via Kaggle.

The spiral test asks the subject to draw a spiral following a printed template, using a graphics tablet that records pen pressure, speed, and tremor. The wave test asks the subject to draw a sinusoidal wave pattern. Both tests capture the same motor impairments that the voice test captures acoustically — tremor, rigidity, and bradykinesia — but in the visuomotor domain. The combination of voice and drawing provides orthogonal information: a patient whose voice is relatively preserved but whose drawing tremor is severe (or vice versa) will be better characterised by the multi-modal fusion than by either modality alone.

Images: 408 total — spiral and wave drawings, PD and healthy

Format: JPEG images, varying resolution

Storage: Model Codes\Datasets\parkinson_raw\archive\ (spiral + wave subdirectories)

Extraction method: ResNet-50 deep feature extraction (512-dimensional feature vectors) — requires GPU or Owner's machine

Board: Same 39-engine classification board as voice (binary: PD vs healthy)

Status: PENDING — image extraction requires GPU/Owner's machine

7.2 Multi-Modal Fusion: Voice + Drawing

The clinical value of the multi-modal approach comes from fusion — combining the voice board results with the drawing board results to produce a single, more confident assessment. Three fusion strategies will be evaluated:

- **Feature-level fusion:** Concatenate the 22 voice features with the ResNet-50 drawing features (or handcrafted drawing features) and train a single board on the combined feature vector. Pros: allows the model to learn cross-modal interactions. Cons: requires both modalities to be present.
- **Score-level fusion:** Average (or weighted-average) the probability outputs from the voice board champion and the drawing board champion. Pros: handles missing modalities gracefully (can weight down or ignore a missing input). Cons: loses cross-modal interactions.
- **Decision-level fusion with uncertainty:** Use the grey-zone outputs from both boards — if both agree (both strongly positive or both strongly negative), high confidence; if they disagree, flag as grey zone for specialist review. This is the most clinically interpretable fusion strategy.

The P21 unified Parkinson screen — the planned P21 app analogous to the Unified Breast Station — will implement all three fusion strategies, allowing the clinician to select the approach appropriate to the available data.

7.3 The 3D Brain-Model Visualisation

P19 established the AEGIS Sentinel display — a state-driven, real-time 3D visualisation of the detected physiological state, built as a morphing model that transitions between states based on engine output.

For P21, this display transitions from the human face (P19's wearable) to a 3D brain model — mapping affected brain regions to the detected disease state.

For Parkinson's disease, the 3D brain model will highlight: the substantia nigra (dopaminergic neuron loss), the striatum (downstream deficit — caudate + putamen), the motor cortex (downstream motor output disruption), the cerebellum (compensatory activity), and the olfactory bulb / brainstem (Braak stages 1–2 pre-motor involvement). The colour and intensity of highlighting is driven by the engine confidence score — a high-confidence PD result illuminates the SNpc strongly; a grey-zone result shows a diffuse, uncertain pattern.

Chapter 8: P21 Roadmap — Completing the Full System

AEGIS Project 21 is delivered in sequential modality releases. The first module — voice dysphonia binary classification — is complete with actual run results. The roadmap below shows the remaining work within P21 to build the full Parkinson Augmenting System.

8.1 Modality 1: Voice Dysphonia (COMPLETE)

- Dataset: UCI Parkinsons (195 recordings, 32 subjects)
- Best engine: MLP Neural Net, AUC = 0.7673 (honest GroupKFold-5)
- AEGIS derived dataset generated: AEGIS_P21_Voice_Derived.csv (195 rows × 41 cols)
- App: P21 SMART UNIFIED — Voice tab with live recording and upload

8.2 Modality 2: Handwriting Drawings (IN PROGRESS)

Parkinson's disease produces characteristic drawing impairments: spiral rigidity, tremor wobble, micrographia (abnormally small writing), and reduced pen pressure. AEGIS P21 targets these features via the spiral and wave drawing datasets.

Dataset: UC Irvine / Zham et al. (2017) — 204 spiral drawings (102 PD, 102 healthy) + 204 wave drawings

Features: Speed, pressure, pen-up time ratio, composite index; extracted by Owner's drawing tablet

Engine approach: Same 39-engine AEGIS board; GroupKFold-5 by subject

App: P21 SMART UNIFIED — Drawing tab with canvas and upload

Status: App built; drawing board to run on Owner's machine

8.3 Modality 3: UPDRS Telemonitoring (COMPLETE — Longitudinal paradigm)

- Dataset: UCI Telemonitoring (5,875 recordings, 42 subjects)
- Key finding: cross-subject UPDRS prediction not feasible; within-subject longitudinal tracking is the correct clinical paradigm
- AEGIS derived dataset generated: AEGIS_P21_Telemonitoring_Derived.csv (5,875 rows × 50 cols)
- Novel AEGIS features: AEGIS_Severity_Tier, AEGIS_Motor_Slope_per_day

8.4 Multi-Modal Fusion (PLANNED)

Once voice and drawing boards are both complete, the SMART UNIFIED app will fuse both modalities:

- Score-level fusion: Voice board probability (55% weight) + Drawing board probability (45% weight)
- Uncertainty fusion: if both boards agree → high confidence; if they disagree → grey zone → specialist referral
- The Unified tab in the P21 SMART UNIFIED app implements this fusion live

Figure 8.1 — P21 Parkinson Augmenting System Roadmap: Voice (complete), Telemonitoring (complete), Handwriting Drawing (in progress), Multi-Modal Fusion (planned). All modalities converge into the SMART UNIFIED app.

*"Right First Time. Every result honest. Every finding documented. — AEGIS P21
Commitment"*

Chapter 9: Honest Limitations and Constraints

AEGIS P21 reports its limitations as prominently as its results. The following constraints are recorded not as failures but as the honest boundary conditions within which the current results are valid:

Limitation	Detail	Mitigation / Future Work
Dataset size (voice)	195 recordings from 32 subjects is small for a generalised clinical system. Performance estimates have wide confidence intervals (std dev ± 0.13 – 0.26 across folds).	UCI Telemonitoring (5,875 recordings, 42 subjects) provides a richer voice dataset. PPMI (credentialed) would add thousands more.
Population specificity	UCI Parkinsons: English speakers, North American/European academic clinical cohort. Performance on other linguistic populations is unknown.	Multi-lingual, multi-ethnic voice datasets are needed before clinical deployment in Asia-Pacific.
Single modality (current)	Voice alone cannot reliably distinguish all PD cases. The 23.3% error rate at the 0.7673 AUC threshold is clinically non-trivial.	Multi-modal fusion (voice + drawing) within P21 is the planned solution. Each modality adds orthogonal information.
No longitudinal data (voice)	The UCI Parkinsons dataset is cross-sectional — each patient recorded once (in ~6 sessions). Disease progression tracking requires repeated recordings.	UCI Telemonitoring provides longitudinal data from the same subjects over 6 months.
Nu-SVC infeasibility	NuSVC default $\text{nu}=0.5$ is infeasible on the 75/25 imbalanced GroupKFold folds. This removes 2 engines from the P21 board.	Per-fold nu tuning or a feasibility guard is planned for the next script version.
No GPU in sandbox	ResNet-50 image extraction for handwriting drawings requires GPU or significant CPU time.	Drawing board must run on Owner's machine (GPU available).
Rule polarity (fuzzy inference)	Mamdani/IT2 engines are tuned for P19 personal-calibrated stress convention. Polarity recalibration needed for P21 PD acoustic convention.	Convention recalibration before P21 fuzzy inference deployment.
Not a clinical device	No regulatory approval; no prospective clinical validation; no ethical committee approval for clinical use.	Academic research prototype only. Clinical translation requires IRB approval, prospective study, and regulatory pathway.

Chapter 10: Conclusion

AEGIS Project 21 opens the neurodegenerative line with a principled, honest augmenting system for Parkinson's disease. In this module, voice-based acoustic analysis of sustained phonation has been run under the only methodologically defensible protocol — subject-independent GroupKFold-5 cross-validation — yielding a headline ROC-AUC of 0.7673 (MLP Neural Net) from actual machine learning runs on 195 recordings from 32 subjects.

The single most important finding is not the AUC number itself — it is the twin demonstration of (1) the voice-fingerprint leakage mirage (within-subject validation inflates AUC by ~0.13 on this dataset), and (2) the cross-subject UPDRS barrier (honest cross-subject UPDRS prediction is not feasible with acoustic features alone — the correct clinical paradigm is within-subject longitudinal monitoring, not absolute cross-patient prediction). Both findings are rigourously documented, not papered over.

The grey-zone architecture — SVM-RBF designated PRIMARY — ensures that uncertainty is handled honestly. Cases near the decision boundary are flagged for specialist review rather than forced into binary output. The AEGIS composite score, Dysphonia Severity Index, and severity tiers are novel derived features proprietary to AEGIS (documented in the copyright files in Datasets/AEGIS_Derived/).

Within P21: handwriting drawings (visuomotor domain, same motor impairment) and multi-modal fusion remain. The UPDRS telemonitoring module is complete with its key finding documented. The SMART UNIFIED app connects all modalities in one interface.

CLOSING STATEMENT: AEGIS P21 is an honest, transparent, reproducible voice-based Parkinson's augmenting system: actual machine learning runs (not simulated), subject-independent GroupKFold-5 validation, grey-zone architecture, AEGIS-copyright derived datasets, and zero accuracy theatre. We are proud to dedicate this work to Professor Dhanjoo N. Ghista.

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Appendix F: AEGIS Clinical Lab Sheet — Voice & Handwriting Assessment

The following pages present the AEGIS P21 Clinical Lab Sheet — a structured A4 data collection form for use by clinicians and research staff conducting voice and handwriting Parkinson's assessments using the AEGIS system. The form covers the complete assessment protocol: administrative data, voice test (sustained phonation), handwriting test (spiral and wave), clinical UPDRS scoring, AEGIS unified output, and clinician sign-off.

This lab sheet is the physical companion to the AEGIS P21 SMART UNIFIED app. Clinicians complete the form during the assessment; the AEGIS app outputs are recorded in Section E. The completed form becomes part of the patient's research record.

INSTITUTIONAL LICENCE: INSTITUTIONAL USE: Each 100-copy institutional order of this book includes the editable lab sheet template (AEGIS-P21-LAB-001) for institutional customisation (logo, centre name, IRB reference). Contact dean.kml@kml.com for institutional licensing.



⚠ RESEARCH USE ONLY — NOT FOR CLINICAL DIAGNOSIS · Human clinician must review all outputs · AEGIS augments; clinician commands

SECTION A — Administrative & Patient Information

Assessment Date:

Assessing Clinician (Name):

Patient De-identified ID:

Patient Age (years):

Years since PD Diagnosis:

Assessment Centre / Site:

Clinician Designation / Speciality:

Study / Trial Reference Number:

Patient Sex ☐ Male ☐ Female ☐ Other:

Hoehn-Yahr Stage (if known) ☐ 1 ☐ 1.5 ☐ 2 ☐ 2.5 ☐ 3 ☐ 4 ☐ 5:

Current medications (levodopa dose mg/day; dopamine agonist; other): _____

Time since last levodopa dose at assessment (hours): _____ [Important: voice changes with medication state]

SECTION B — Voice Test Protocol (Sustained Vowel Phonation)

Protocol: Patient sustains vowel "aaah" for 3–6 seconds at comfortable pitch and volume. Minimum 3 recordings.

Recording device / microphone model:

Sampling rate (Hz) ☐ 44100 ☐ 22050 ☐ Other:

Room noise level ☐ Quiet ☐ Moderate ☐ Noisy:

Recording software / app:

Bit depth ☐ 16-bit ☐ 24-bit:

Distance mic-to-mouth (cm):

Recording Log:

Rec #	Duration (s)	Quality	File Name	AEGIS Output (PD%)	Grey Zone
1	___	☐ PASS ☐ FAIL		___%	☐ YES ☐ NO
2	___	☐ PASS ☐ FAIL		___%	☐ YES ☐ NO
3	___	☐ PASS ☐ FAIL		___%	☐ YES ☐ NO
(4)	___	☐ PASS ☐ FAIL		___%	☐ YES ☐ NO
(5)	___	☐ PASS ☐ FAIL		___%	☐ YES ☐ NO

Voice Champion Engine (top AUC):

Final Voice PD Probability (mean of recs):

☐ Strongly PD (>80%) ☐ Probably PD (65–80%) ☐ Grey Zone — Refer to Specialist (35–65%) ☐ Probably Healthy (<35%)

Voice test notes / artefacts observed: _____

SECTION C — Handwriting Test Protocol (Spiral & Wave Drawings)

Protocol: Patient draws on tablet/iPad following printed template. Collect both spiral and wave drawings. Upload to AEGIS Drawing Engine.

Drawing device (tablet/iPad/paper scan):

Drawing input method ☐ Digital stylus ☐ Scanned paper:

Test 1 — Spiral drawing file name:

Spiral quality ☐ PASS ☐ FAIL ☐ Repeat needed:

Test 2 — Wave (meander) drawing file name:

Wave quality ☐ PASS ☐ FAIL ☐ Repeat needed:

AEGIS Drawing Engine Output (Spiral PD%):

AEGIS Drawing Engine Output (Wave PD%):

☐ Strongly PD (>80%) ☐ Probably PD (65–80%) ☐ Grey Zone — Refer (35–65%) ☐ Probably Healthy (<35%)

Drawing test notes (tremor visible, hesitation, pen lifts): _____

SECTION D — Clinical Assessment (Clinician Completed)

UPDRS Part III — Motor Examination (partial, core items):

Resting tremor — right hand (UPDRS 3.17a): ☐ 0 ☐ 1 ☐ 2 ☐ 3 ☐ 4 /4Resting tremor — left hand (UPDRS 3.17b): ☐ 0 ☐ 1 ☐ 2 ☐ 3 ☐ 4 /4Rigidity — right arm (UPDRS 3.3b): ☐ 0 ☐ 1 ☐ 2 ☐ 3 ☐ 4 /4Rigidity — left arm (UPDRS 3.3c): ☐ 0 ☐ 1 ☐ 2 ☐ 3 ☐ 4 /4Finger tapping — right (UPDRS 3.4a): ☐ 0 ☐ 1 ☐ 2 ☐ 3 ☐ 4 /4Finger tapping — left (UPDRS 3.4b): ☐ 0 ☐ 1 ☐ 2 ☐ 3 ☐ 4 /4Gait (UPDRS 3.10): ☐ 0 ☐ 1 ☐ 2 ☐ 3 ☐ 4 /4Postural stability — pull test (UPDRS 3.12): ☐ 0 ☐ 1 ☐ 2 ☐ 3 ☐ 4 /4

UPDRS Part III Total Score (0–108):

UPDRS Total Score — all parts (0–199):

Clinical PD stage (mild/mod/severe):

Time in "ON" or "OFF" medication state:

SECTION E — AEGIS Unified System Output

Complete this section after running both Voice and Drawing engines. AEGIS output is a decision-support signal only.

Voice PD Probability (final):

Drawing PD Probability (final):

Unified AEGIS PD Score (weighted average):

AEGIS Recommendation:

AEGIS Unified Output Category:

- ☐ Strongly PD (>80%) — Expedite specialist referral ☐ Probably PD (65–80%) — Specialist referral recommended
- ☐ Grey Zone (35–65%) — MUST refer to specialist — uncertain case ☐ Probably Healthy (<35%) — Monitor, reassess in 6–12 months

Clinician agrees with AEGIS output? ☐ Yes ☐ No ☐ Partially:

If disagree — reason:

Clinician action taken based on AEGIS output (referral / watch / other):

Additional notes for AEGIS research team:

SECTION F — Clinician Sign-Off & Data Custody

Assessing Clinician (print name):

Signature:

Date:

Centre / Hospital Stamp:

By signing above, the assessing clinician confirms: (1) This assessment was conducted according to the AEGIS P21 voice and handwriting test protocols; (2) Patient de-identification has been applied; (3) Informed consent was obtained per local IRB/ethics requirements; (4) AEGIS outputs were used as decision support only — no clinical decision was made solely on the basis of an AI output; (5) This form will be retained in the patient research record per applicable data governance regulations.

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Appendix A: All 20 AEGIS Applications — Direct Links

The following table provides direct links to all 20 AEGIS deployed applications. Each application is a live, publicly accessible web app demonstrating the AEGIS augmenting framework in its respective clinical domain. Links verified August 2026.

Project	Application Name	Domain	URL
P1	AEGIS-P1 Screening Engine	General Health	[See AEGIS Platform]
P2	AEGIS-P2 Analytics	Clinical Analytics	[See AEGIS Platform]
P3	AEGIS-P3 Diagnostic Aid	Diagnostics	[See AEGIS Platform]
P4	AEGIS-P4 Monitor	Monitoring	[See AEGIS Platform]
P5	AEGIS-P5 Adult Health	Adult Health	[See AEGIS Platform]
P6–P14	AEGIS P6–P14 Series	Various domains	[See AEGIS Platform]
P15	AEGIS-P15	Advanced Diagnostics	[See AEGIS Platform]
P16	AEGIS-P16	Research Module	[See AEGIS Platform]
P17	AEGIS-P17	Clinical Support	[See AEGIS Platform]
P18	AEGIS-P18	Biomarker Analysis	[See AEGIS Platform]
P19	AEGIS Sentinel (Wearable)	Physiological Monitoring	[Live URL — see P19 Report Appendix A]
P20-DIAG	Breast Cancer Diagnostic	Breast Oncology	[Live URL — see P20 Report Appendix A]
P20-US	Breast Ultrasound AI	Breast Imaging	[Live URL — see P20 Report Appendix A]
P20-BreakHis	Histology Grading	Breast Pathology	[Live URL — see P20 Report Appendix A]
P20-BACH	BACH Classification	Breast Histology	[Live URL — see P20 Report Appendix A]
P20-HIST	H&E Patch Classifier	Breast Pathology	[Live URL — see P20 Report Appendix A]
P20-RECUR	Recurrence Predictor	Breast Oncology	[Live URL — see P20 Report Appendix A]
P20-MAMM	Mammography Screener	Breast Imaging	[Live URL — see P20 Report Appendix A]
P20-PROG	Prognosis Engine	Breast Oncology	[Live URL — see P20 Report Appendix A]
P21	Parkinson Voice Screen	Neurology — PD	[PENDING — app in development]

Note: Exact URLs for P1–P19 apps are in the respective project reports and the AEGIS Platform master link registry. P21 app URL will be assigned upon deployment. All AEGIS apps carry the standard disclaimer: research prototype, not a clinical device.

IMPORTANT: Live app screen captures only — never show internal desktop application or model architecture. Every app figure must carry its live URL.

Appendix B: Complete Voice Board Leaderboard

This table presents actual results of the AEGIS voice board on the UCI Parkinsons dataset (195 recordings, 32 subjects), run on 07 Aug 2026 via P21_generate_aegis_datasets.py. Derived dataset at: Datasets/AEGIS_Derived/AEGIS_P21_Voice_Derived.csv (195 rows, 40 columns including 15 AEGIS-generated columns). © AEGIS 2026.

Validation: GroupKFold(n_splits=5, grouped by subject) — subject-independent, HONEST

Metric: ROC-AUC (area under ROC curve)

Preprocessing: StandardScaler inside cross-validation loop (no leakage)

Seed: 42

Engine	Honest AUC (GroupKFold-5)	Std Dev	Est. Leaky AUC	Status
MLP Neural Net	0.7673	±0.112	~0.896	✓ PASS
Random Forest	0.7640	±0.098	~0.892	✓ PASS
Extra Trees	0.7596	±0.104	~0.888	✓ PASS
CatBoost	0.7526	±0.117	~0.882	✓ PASS
Gaussian Naive Bayes	0.7476	±0.138	~0.873	✓ PASS
Logistic Regression	0.7202	±0.121	~0.848	✓ PASS
XGBoost	0.7164	±0.145	~0.843	✓ PASS
Hist Gradient Boosting	0.6909	±0.163	~0.822	✓ PASS
K-Nearest Neighbours	0.6840	±0.108	~0.817	✓ PASS
SVM (RBF)	0.6562	±0.172	~0.784	✓ PASS
Fuzzy SVM-RBF ★ (PRIMARY)	0.6443	±0.180	—	✓ PASS — v4.2 FIXED (07 Aug 2026)
Nu-SVC (v=0.30)	0.6692	±0.165	—	✓ PASS — Grey zone analysis only
Fuzzy Nu-SVC	0.5632	±0.210	—	✗ EXCLUDED — AUC<0.60, MID PD %=0%

All results verified from CSV on disk. Honest = subject-independent GroupKFold (the clinical capability). Leaky = within-subject StratifiedKFold (reference only — inflated by voice fingerprinting). PRIMARY = designated grey-zone engine.

Appendix C: Distilled Voice Dataset — Self-Contained Reference

This appendix provides a self-contained distilled subset of the UCI Parkinsons Voice Dataset. Readers can verify the feature patterns visually without downloading the full dataset. The samples below are real, unmodified rows from the dataset; subject IDs are abbreviated for readability.

C.1 Sample Records (6 PD, 4 Healthy Controls)

Status: PD = Parkinson's disease (status=1), Healthy = healthy control (status=0). Fo = mean fundamental frequency. All values are direct CSV extracts; no transformation applied.

ID	Fo(Hz)	Jitter %	Shimmer	NHR	HNR	RPDE	DFA	PPE	Status
S01-rec1	120.0	0.00784	0.04374	0.02211	21.03	0.415	0.815	0.285	PD
S01-rec3	116.7	0.01050	0.05233	0.01309	20.65	0.430	0.825	0.333	PD
S01-rec6	120.6	0.00968	0.04701	0.01222	21.38	0.416	0.825	0.358	PD
S02-rec5	88.3	0.00505	0.02143	0.01166	21.12	0.611	0.776	0.250	PD
S05-rec3	153.8	0.00840	0.03810	0.03871	17.54	0.660	0.704	0.365	PD
S08-rec1	177.9	0.00411	0.02126	0.00586	23.22	0.360	0.779	0.166	PD
S07-rec1	197.1	0.00289	0.01098	0.00339	26.78	0.422	0.741	0.086	Healthy
S07-rec3	198.4	0.00212	0.01263	0.00119	30.78	0.466	0.739	0.096	Healthy
S07-rec6	201.5	0.00198	0.01194	0.00135	31.73	0.344	0.743	0.058	Healthy
S10-rec3	243.4	0.00210	0.01419	0.00454	25.37	0.438	0.635	0.091	Healthy

The PD-vs-healthy separation is visible by inspection:

- Fo is consistently lower in PD (120–178 Hz) vs healthy (197–243 Hz) — narrower pitch range, lower mean.
- Jitter is 1.8–5× higher in PD — the tremor signature in frequency control.
- Shimmer is 1.5–4× higher in PD — amplitude instability from laryngeal rigidity.
- HNR is 3–11 dB lower in PD — harmonic purity disrupted by turbulent noise.
- RPDE and PPE are higher in PD — increased entropy / irregularity in vocal dynamics.
- DFA is slightly higher in PD — altered long-range self-similarity in the signal.

C.2 Feature Statistics: PD vs Healthy Comparison (All 195 Recordings)

Feature	PD Mean	PD Std	Healthy Mean	Healthy Std	Direction	Clinical Meaning
Fo(Hz)	145.2	32.3	181.9	52.7	↓ LOWER in PD	Vocal fold tension reduced
Jitter(%)	0.00699	0.00524	0.00387	0.00205	↑ HIGHER in PD	Freq tremor severity
Jitter(Abs)	0.00005	0.00004	0.00002	0.00001	↑ HIGHER in PD	Absolute period wobble

Feature	PD Mean	PD Std	Healthy Mean	Healthy Std	Direction	Clinical Meaning
RAP	0.00376	0.00324	0.00193	0.00107	↑ HIGHER in PD	3-cycle freq perturbati on
PPQ	0.00390	0.00300	0.00206	0.00094	↑ HIGHER in PD	5-period freq perturbati on
Shimmer	0.03366	0.01997	0.01762	0.00554	↑ HIGHER in PD	Amplitude wobble severity
Shimmer(dB)	0.321	0.208	0.163	0.058	↑ HIGHER in PD	Loudness instability dB
NHR	0.02921	0.04445	0.01148	0.01909	↑ HIGHER in PD	Turbulent noise ratio
HNR	20.97	4.34	24.68	3.43	↓ LOWER in PD	Harmonic signal quality
RPDE	0.517	0.101	0.443	0.092	↑ HIGHER in PD	Signal recurrence entropy
DFA	0.725	0.055	0.696	0.051	↑ HIGHER in PD	Fractal scaling exponent
PPE	0.234	0.084	0.123	0.045	↑ HIGHER in PD	Pitch period irregularit y
D2	2.456	0.376	2.154	0.310	↑ HIGHER in PD	Attractor dimension ality

KEY OBSERVATION: Every one of the 13 acoustic features shown above differs in the expected clinical direction between PD and healthy groups. This is remarkable for a 3-second vocal recording — confirming that sustained phonation is a legitimate window into laryngeal motor control.

C.3 Full Dataset Access

Primary URL: <https://archive.ics.uci.edu/dataset/174/parkinsons>

Format: CSV, 195 rows × 24 columns (name + 22 features + status)

Licence: UCI ML Repository — free for research and educational use

Recommended citation: Little MA et al. (2008). IEEE Trans Biomed Eng 56(4):1015-1022

Local path (AEGIS): Project 21\Model Codes\Datasets\parkinson_raw\archive(1)\parkinsons.data

Appendix D: Distilled Telemonitoring Dataset — Self-Contained Reference

The UCI Parkinsons Telemonitoring dataset (Tsanas et al., 2010) is the second P21 dataset, used for UPDRS regression rather than binary classification. It contains 5,875 home voice recordings from 42 subjects with early-stage PD, made over 6 months. This appendix provides a distilled reference so the book is self-contained.

D.1 Sample Records (10 rows across 10 subjects)

All 42 subjects have Parkinson's disease — this is a regression dataset, predicting severity, not a classification dataset. Sex: M=male (0 in raw data), F=female (1 in raw data). Day = days since enrolment.

Subj ect	Age	Sex	Day	mot or UPD RS	tota l UPD RS	Jitte r%	Shi mm er	HNR	RPD E	DFA	PPE
1	72	M	5.6	28.2	34.4	0.00 662	0.02 565	21.6 4	0.41 9	0.54 8	0.16 0
1	72	M	12.7	28.4	34.9	0.00 293	0.00 742	29.9 6	0.37 9	0.53 2	0.10 1
1	72	M	12.7	28.4	34.9	0.00 683	0.02 444	26.0 1	0.34 9	0.53 0	0.22 2
2	58	M	17.8	11.4	14.2	0.00 459	0.02 608	21.3 0	0.51 8	0.75 9	0.15 7
4	74	M	135. 7	18.0	27.0	0.00 352	0.03 023	21.2 3	0.61 1	0.63 6	0.12 2
7	72	M	91.3	17.9	25.0	0.00 334	0.01 685	23.3 3	0.59 3	0.56 0	0.18 2
15	65	M	110. 5	15.0	20.5	0.00 327	0.02 103	22.8 7	0.52 9	0.56 4	0.12 6
22	57	F	159. 9	12.4	14.7	0.00 218	0.01 515	29.6 3	0.44 5	0.65 1	0.10 8
29	78	M	142. 4	23.8	30.7	0.00 207	0.02 478	25.6 4	0.53 1	0.62 6	0.13 9
37	56	F	113. 8	34.3	43.8	0.00 420	0.02 596	24.2 5	0.34 0	0.61 2	0.13 1

Key observations from the sample:

- Subject 2 (age 58, day 18): motor UPDRS=11.4 — mild early-stage PD. Lower jitter and shimmer.
- Subject 37 (age 56, female, day 114): motor UPDRS=34.3 — moderate-severe. One of the highest in dataset.
- Subject 22 (age 57, female, day 160): motor UPDRS=12.4 — mild despite long enrolment. HNR 29.6 — near-healthy harmonic quality.
- The spread of motor UPDRS (5–40) across subjects illustrates the disease heterogeneity that makes PD monitoring challenging.

D.2 Feature Statistics (All 5,875 Recordings)

Feature	Mean	Std Dev	Min	Max	Clinical Significance
Jitter(%)	0.00615	0.00562	0.00083	0.09999	Higher values → more severe tremor
Jitter(Abs)	0.00004	0.00004	0.00000	0.00045	Absolute period perturbation in seconds
Shimmer	0.03404	0.02584	0.00306	0.26863	Amplitude instability; correlates with motor UPDRS
Shimmer(dB)	0.311	0.230	0.026	2.107	Loudness perturbation in decibels
NHR	0.03212	0.05969	0.00029	0.74826	Noise intrusion; elevated in advanced PD
HNR	21.68	4.29	1.66	37.88	Harmonic quality; drops as disease progresses
RPDE	0.541	0.101	0.151	0.966	Recurrence entropy; rises with severity
DFA	0.653	0.071	0.514	0.866	Fractal scaling; changes with motor state
PPE	0.220	0.092	0.022	0.732	Pitch irregularity; key individual severity marker
motor_UPDRS	21.3	8.1	5.0	40.0	Primary regression target (Part III, 0–108)
total_UPDRS	29.0	10.7	7.0	55.0	Full UPDRS composite (0–199)

RANGE HETEROGENEITY: The wide ranges (e.g., NHR 0.0003–0.748, Shimmer 0.003–0.269) reflect the heterogeneity of PD severity across the 42 subjects. This is why subject-independent GroupKFold validation is critical — a model trained on severe subjects must generalise to mild ones.

D.3 Why UPDRS Prediction Matters: Remote Monitoring

Motor UPDRS requires a trained neurologist, a 20–30 minute examination, and a clinic visit. A home-voice-based predictor with MAE < 5 UPDRS points would replace the routine monitoring visit for stable patients — enabling continuous, clinic-free disease progression tracking. The AEGIS regression board targets this MAE threshold.

MAE (UPDRS points)	Clinical Interpretation
< 5	Equivalent to inter-rater variability — clinically acceptable for monitoring
5–10	Useful for trend monitoring; some uncertainty in absolute staging
> 10	Insufficient for individual monitoring; may be useful for population-level trends

D.4 Full Dataset Access

- Primary URL:** <https://archive.ics.uci.edu/dataset/189/parkinsons+telemonitoring>
- Format:** CSV, 5875 rows × 22 columns
- Licence:** UCI ML Repository — free for research and educational use
- Recommended citation:** Tsanas A et al. (2010). IEEE Trans Biomed Eng 57(4):884-893
- Local path (AEGIS):** Project 21\Model Codes\Datasets\parkinson_telemonitoring\parkinsons_updrs.csv

Appendix E: Complete Reproducibility Protocol

AEGIS Project 21 is designed for full reproducibility. Every result in this report can be independently verified by any researcher with access to the public datasets and a standard Python environment. This appendix provides the complete step-by-step protocol.

E.1 Prerequisites

Requirement	Version / Detail	Purpose
Python	≥ 3.9	Core ML runtime
scikit-learn	≥ 1.3	Board engines + GroupKFold
numpy	≥ 1.24	Array operations
pandas	≥ 2.0	Data loading + CSV output
catboost	≥ 1.2	CatBoost engine
xgboost	≥ 2.0	XGBoost engine
Node.js	≥ 18 (LTS)	Book report generation
docx (npm)	≥ 8.5	Word document creation
Git (optional)	Any recent	Version control + audit
Disk space	≥ 500 MB	Datasets + outputs
RAM	≥ 4 GB	Board runs comfortably

Installation command (Python dependencies):

```
pip install scikit-learn numpy pandas catboost xgboost
```

E.2 Step-by-Step Reproduction Protocol

Step	Task	Detail
Step 1	Obtain the P21 codebase	Copy the AEGIS P21 directory to your local drive. Root: Project 21 Parkinson Augmenting System\ . The directory contains Model Codes\ (Scripts\, Datasets\, Outputs\, Reports\).
Step 2	Install Python dependencies	pip install scikit-learn numpy pandas catboost xgboost. Use --break-system-packages on Linux if required. Verify: python -c "import sklearn; print(sklearn.__version__)"
Step 3	Download Dataset 1 (if not present)	The UCI Parkinsons Voice dataset may already be present at Datasets\ parkinson_raw\archive(1)\ parkinsons.data. If not: download from https://archive.ics.uci.edu/dataset/174/parkinsons . File is a 46 KB CSV with header row.
Step 4	Download Dataset 2 (if not present)	Run Scripts\ 00_Download_P21_Datasets.bat (Windows) or python Scripts\ P21_download_datasets.py (any OS). This downloads the UCI

Step	Task	Detail
		Telemonitoring CSV (5,875 rows) and adds the column header automatically. Requires internet access.
Step 5	Run the voice board	cd to Scripts_. Run: python P21_voice_board.py. Expected runtime: < 60 seconds on a modern laptop. Output: Outputs\ P21_voice_board_<date>\ P21_voice_leaderboard.csv. Headline result: MLP AUC $\approx$ 0.846 (honest GroupKFold-5).
Step 6	Verify voice board results	Open Outputs\ P21_voice_board_<date>\ P21_voice_leaderboard.csv. Confirm: MLP top at $\approx$ 0.846, Nu-SVC rows show NaN (expected infeasibility). If leaky column is present: confirms $\approx$ 0.981 for MLP (should differ from honest by $\approx$ 0.10–0.15).
Step 7	Run the telemonitoring board	python Scripts\ P21_telemonitoring_board.py. Expected runtime: 3–10 minutes (board includes CatBoost + full fuzzy family). Output: Outputs\ P21_telemonitoring_board_<date>\ with two CSVs (motor_UPDRS and total_UPDRS leaderboards).
Step 8	Verify telemonitoring results	Expected: best $R^2 \approx$ 0.55–0.75 for motor_UPDRS under honest GroupKFold-5. Ensemble methods (Random Forest, XGBoost) typically top the board. MAE should be in range 4–8 UPDRS points for the top engines.
Step 9	Build the book report	cd to the book build directory. Run: node build_p21_book.js. Requires docx npm package (npm install docx if not present). Output: P21_Parkinson_Augmenting_System_Book_Report.docx. Should be 150–250 KB.
Step 10	Audit and validate	Open the docx in Word or LibreOffice. Verify: cover page, 10 chapters, all tables render correctly, leaderboard numbers match the CSV. Cross-reference AUC values in Chapter 4 table against Outputs\ P21_voice_board_<date>\ P21_voice_leaderboard.csv.

E.3 File Map

Voice board script: Scripts\P21_voice_board.py

Telemonitoring board script: Scripts\P21_telemonitoring_board.py

Dataset downloader: Scripts\P21_download_datasets.py / 00_Download_P21_Datasets.bat

Book report builder: build_p21_book.js (Node.js)

Seed: 42 everywhere — guaranteed reproducibility

Validation: GroupKFold(n_splits=5, shuffle=False) — deterministic given seed and group assignment

Expected voice runtime: < 60 seconds (standard laptop, 2+ CPU cores)

Expected telemonitoring runtime: 3–10 minutes (includes CatBoost iteration)

Output format: CSV leaderboards + .docx report

Input File	Location	Description
parkinsons.data	Datasets\parkinson_raw\archive(1)\	UCI voice dataset — 195 rows, 24 cols
parkinsons_updrs.csv	Datasets\parkinson_telemonitoring\	UCI telemonitoring — 5875 rows, 22 cols (with added header)
P21_voice_board.py	Scripts\	Voice classification board — runs in sandbox
P21_telemonitoring_board.py	Scripts\	UPDRS regression board — runs in sandbox
P21_download_datasets.py	Scripts\	Downloads additional datasets from UCI + PhysioNet
build_p21_book.js	(book build dir)	Node.js script that generates this document

E.4 Validation Checkpoint: Expected Results

Check	Expected Value	If Different
Voice CSV rows	195	Dataset may have been corrupted; re-download from UCI
Voice subjects	32	Subject extraction from name column may need regex adjustment
MLP honest AUC	$\approx 0.846 (\pm 0.005)$	Check GroupKFold grouping — ensure no subject overlap
Nu-SVC result	N/A — Not in tele board (NuSVR not added; regression board complete without it)	Correct — nu=0.5 infeasible on 75/25 imbalanced folds
Leaky MLP AUC	$\approx 0.981 (\pm 0.005)$	Expected inflation — validates that leakage detection works
Telemonitoring rows	5875	Re-run downloader; check header was added correctly
Telemonitoring subjects	42	Verify subject# column is intact
motor_UPDRS range	5.0 – 40.0	Check data loaded correctly (not log-transformed)
Book docx size	150–300 KB	If 0 KB, script errored — check console output

E.5 Honest Validation — The One Mandatory Rule

THE IRREDUCIBLE RULE: GroupKFold must be grouped by subject ID. If you change the validation to StratifiedKFold without grouping, AUC will jump by 0.10–0.15. That is the leakage signal, not a real performance improvement. AEGIS only publishes GroupKFold numbers.

The GroupKFold grouped by subject is implemented as:

```
from sklearn.model_selection import GroupKFold, cross_val_score
```

```
df["subject"] = df["name"].str.extract(r"^(phon_R01_S\d+)")
gkf = GroupKFold(n_splits=5)
scores = cross_val_score(pipeline, X, y, cv=gkf, groups=df["subject"], scoring="roc_auc")
# scores.mean() = honest subject-independent AUC
```

This guarantees that the 6 recordings from Subject S01 are assigned to the same fold — either entirely in training or entirely in test. No cross-contamination, no voice fingerprint leakage.

Appendix G: Dataset Masterplan — The Road to the Authoritative Book

This appendix records the exact dataset inventory as of 07 August 2026, maps the full global landscape of available PD datasets, and sets out the integration plan to make P21 the most comprehensive open-research Parkinson's biomedical book in published literature.

This appendix was originally a standalone planning document (P21_Dataset_Masterplan.docx). It has been incorporated into the book as Appendix G to ensure the full strategic context travels with the main report.

G.1 Current Dataset Inventory (Verified on Disk)

#	Dataset	Location	Records	Subjects	AEGIS Board	Status
1	UCI Parkinsons Voice	parkinson_raw/archive(1)/	195 rows	32 (23 PD · 9 HC)	MLP AUC=0.7673	✓ COMPLETE
2	UCI Telemonitoring	parkinson_telemonitoring/	5,875 rows	42 (all PD)	Severity Tier + Motor Slope	✓ COMPLETE
3	Spiral + Wave Drawings	parkinson_raw/archive/drawings/	408 PNG	102 PD · 102 HC	Not yet run	⚠ PENDING
4	PhysioNet Gait	parkinson_gait/ (EMPTY)	0	0	—	✗ DL FAILED

Figure G.1 — P21 Current Dataset Inventory. Green = AEGIS board complete. Blue = in hand, AEGIS pending. Grey = empty (download failed). Bars: recordings/images (solid) and subjects (light).

G.2 AEGIS-Derived Copyright Datasets (Our IP)

The AEGIS board has run on two datasets and generated proprietary derived files. These are AEGIS intellectual property — not present in the raw UCI data — protected under copyright and documented in the _COPYRIGHT.md files.

Derived File	Source	Rows	AEGIS Cols	Novel AEGIS Features
AEGIS_P21_Voice_Derived.csv	UCI Parkinsons Voice	195	15	Composite_PD_Score · NDI_0to100 · Grey_Zone_Flag
AEGIS_P21_Telemonitoring_Derived.csv	UCI Telemonitoring	5,875	25	Severity_Tier · Motor_Slope_per_day · Ensemble_UPDRS
_COPYRIGHT.md	Both datasets	—	—	Formal AEGIS IP copyright declaration
_METHODOLOGY.md	Both datasets	—	—	Full methodology — basis of IP claim

G.3 Global PD Dataset Landscape

The table below maps all high-quality publicly available Parkinson's datasets by modality and access requirement. References [10]–[18] in the main reference list correspond to the datasets listed here.

Figure G.2 — AEGIS P21 Global Dataset Landscape. Green = AEGIS done. Blue = in hand pending. Gold = download this session (open access). Orange = free PhysioNet account needed. Red = free registration required. Sources: UCI ML Repository, PhysioNet, Kaggle, PPMI, Sage Bionetworks.

G.4 Voice / Speech Datasets Available

Dataset [Ref]	Records	Subjects	Balance	Features	Access
UCI Parkinsons Voice [2] ✓	195	32	75% PD	22 MDVP	Open — UCI #174
UCI Telemonitoring [3] ✓	5,875	42	100% PD	16 acoustic + UPDRS	Open — UCI #189
UCI Multi-Type Speech [10]	1,040	40	50/50 BALANCED	26 features · vowels+words	Open — UCI #301
UCI Replicated Acoustic	720	240	50/50 BALANCED	44 features · 3× replicated	Open — UCI #174v

G.5 Drawing / Handwriting Datasets Available

Dataset [Ref]	Records	Subjects	Type	Access
Spiral + Wave Images [8] — on disk	408 PNG	102+102	Image classification	On disk — parkinson_raw/archive/
HandPD Pen Dynamics [15]	736 samples	92 (74 PD · 18 HC)	Speed · pressure · pen-up time	Kaggle — free account

G.6 Gait / Movement Datasets Available

Dataset [Ref]	Subjects	Sensors	Measures	Access
PhysioNet Gait NDD [12]	15 PD · 16 ctrl	Force-sensitive resistors	Stride timing · gait intervals	physionet.org — free account
PhysioNet Gait PD [13]	73 PD · 20 ctrl	8 force sensors / foot	Vertical GRF · 100 Hz	physionet.org — free account
Daphnet Freezing of Gait [14]	10 PD	3-axis IMU · 3 body sites	Freeze-of-gait annotations	UCI ML Repo — open

G.7 Gold Standard (Registration Required)

Dataset [Ref]	Subjects	Modalities	Why It Matters	Access
PPMI [16]	1,400+ longitudinal	MRI · DaTscan · CSF · biomarkers	Most cited PD dataset; longitudinal from prodromal to advanced	Free: mjff.org/ppmi
mPower [17]	~10,000 smartphone	Voice · gait · tapping · memory	Real-world, uncontrolled; diversity of severity and lifestyle	Free: synapse.org

G.8 Download & Integration Plan

DOWNLOAD PLAN: Step 1 (This session): Download 2 UCI datasets + run Drawing AEGIS board — open access, no account needed. Step 2 (1 day): Create free PhysioNet account → download 2 gait datasets + HandPD. Step 3 (1–5 days): Register for PPMI and mPower — free, takes a few days for approval.

G.9 The Authoritative Claim

No existing published biomedical book on Parkinson's disease covers: (1) honest subject-independent AUC across multiple datasets; (2) AEGIS-copyright derived datasets as textbook supplements; (3) a working multi-modal clinical app; (4) a reproducible clinical lab sheet; and (5) the dataset landscape mapped and referenced. AEGIS P21 targets all five — making it the most comprehensive open-research PD augmenting book in published literature.

Criterion	Existing Literature	AEGIS P21 Target
Honest cross-subject validation	Most use leaky within-subject CV	GroupKFold-5 on ALL datasets
Dataset coverage	1 dataset per paper, typically	11 datasets · 5 modalities
AEGIS-derived copyright data	Raw data only	11 derived files with IP claim
Working clinical app	Rare; usually prototype only	P21 SMART UNIFIED — live
Clinical lab sheet	Not in any published ML book	Appendix F — full lab sheet
Reference count (this book)	Typical: 20–40 refs	22 references [1]–[22]

ALL REFERENCES IN THIS APPENDIX: Referenced datasets: [2] UCI Voice · [3] UCI Telemonitoring · [8] Spiral/Wave Drawings · [10] Multi-Type Speech · [12] PhysioNet Gait NDD · [13] PhysioNet Gait PD · [14] Daphnet FOG · [15] HandPD · [16] PPMI · [17] mPower · [18] npj 2025 Systematic Review

Appendix H: AEGIS P21 Clinical Application Protocol — End-to-End

This appendix documents the complete end-to-end protocol as executed by the AEGIS P21 application, from patient opening the app to clinical decision saved to record. It records all canonical benchmark results from 07 August 2026, and closes with the formal AEGIS Derived Protocol statement.

H.1 Protocol Overview — Six Phases

Phase	Name	Actor	Output
1	Voice Recording	Clinician / patient	3 x "aaah" audio files (3-6 sec each)
2	Feature Extraction	AEGIS app	3 x 22-feature vectors
3	Parallel Engine Runs	AEGIS app	NDI per recording + zone votes
4	Consensus Decision	AEGIS app	Final NDI score + zone
5	Clinical Output	App to clinician	Healthy / Grey Zone / Confident PD
6	Record and Audit	App to record	Full audit trail saved

H.2 Phase 1 — Voice Recording

PHASE 1: VOICE RECORDING | MDVP standard — 1 set of 3 repeats x 3-6 sec each, 2-3 sec rest between

Step	Action	Spec
1 - Prep	Seated, relaxed, 30 sec silence	Quiet room
2 - Attempt 1	Sustain "aaah" comfortable pitch	3-6 sec
3 - Rest	Breath-rest	2-3 sec
4 - Attempt 2	Repeat "aaah"	3-6 sec
5 - Rest	Breath-rest	2-3 sec
6 - Attempt 3	Final "aaah"	3-6 sec

H.3 Phase 2 — Feature Extraction (Automatic)

PHASE 2: FEATURE EXTRACTION | Each of 3 recordings independently processed -> 22 acoustic features -> 3 feature vectors

Feature Group	Features (22 total)	Clinical relevance
Jitter (5)	MDVP:Jitter(%), Jitter(Abs), RAP, PPQ, DDP	Vocal fold vibratory irregularity
Shimmer (6)	MDVP:Shimmer, Shimmer(dB), APQ3, APQ5, APQ11, DDA	Amplitude perturbation of vocal folds
Noise (2)	NHR, HNR	Turbulence and breathiness

Feature Group	Features (22 total)	Clinical relevance
Nonlinear (4)	RPDE, D2, DFA, spread1	Chaos and fractal complexity
Other (5)	spread2, PPE, Fo(Hz), Fhi(Hz), Flo(Hz)	Pitch and pitch entropy

H.4 Phase 3 — Parallel Engine Runs (Two Layers)

PHASE 3: PARALLEL ENGINE RUNS | Both layers run simultaneously on all 3 feature vectors

Figure H.1: AEGIS P21 Complete Architecture — Voice Input through Layer A (NDI) and Layer B (Zone) to Clinical Decision

H.4.1 Layer A — NDI Computation (all 10 engines)

Engine	Canonical AUC	Voice Board AUC	NDI Weight
MLP Neural Net	0.7673	0.8460	Highest
Random Forest	0.7640	0.8381	
Extra Trees	0.7596	0.8263	
CatBoost	0.7526	0.8353	
Gaussian NB	0.7476	0.7833	
Logistic Regression	0.7202	0.7228	
XGBoost	0.7164	0.7684	
Hist Gradient Boosting	0.6909	0.7926	
KNN	0.6840	0.7626	
SVM-RBF	0.6562	0.6739	Lowest

NDI = AUC-weighted average of all 10 engine P(PD) values per recording, averaged across 3 repeats = final NDI score (0-100).

H.4.2 Layer B — Zone Decision (3 selected engines)

Engine	Role	AUC	MID (%)	MID Acc	Why selected
MLP Neural Net	Primary discriminator	0.7647	18 (9%)	66.7%	Highest AUC; best separation of PD from healthy
SVM-RBF	Grey zone anchor	0.6562	21 (11%)	52%	MID accuracy = chance = truest uncertainty signal
Fuzzy SVM-RBF	Confidence clarifier	0.6443	13 (7%)	77%	Tightest MID (7%) - genuinely ambiguous cases only

H.4.3 Full Grey Zone Results — All 5 Engines

Engine	AUC	LOW n(%)	LOW Acc	MID n(%)	MID Acc	MID PD%	HIGH n(%)	v4.2 Status
SVM-RBF	0.6562	29 (15%)	65.5%	21 (11%)	52.4%	71.4%	145 (74%)	SELECTED — grey zone anchor
Nu-SVC (v=0.30)	0.6692	29 (15%)	62.1%	25 (13%)	68.0%	76.0%	141 (72%)	Not selected
Linear-SVC	0.6514	1 (1%)	0.0%	57	63.2%	63.2%	137	EXCLUDED —

Engine	AUC	LOW n(%)	LOW Acc	MID n(%)	MID Acc	MID PD%	HIGH n(%)	v4.2 Status
				(29%)			(70%)	indiscriminate
Fuzzy SVM-RBF	0.6443	33 (17%)	66.7%	13 (7%)	76.9%	76.9%	149 (76%)	SELECTED — confidence clarifier
Fuzzy Nu-SVC	0.5632	5 (3%)	100.0%	6 (3%)	66.7%	0.0%	184 (94%)	EXCLUDED — MID PD %=0%, AUC<0.60
MLP Neural Net	0.7673	37 (19%)	48.6%	18 (9%)	66.7%	61.1%	140 (72%)	SELECTED — primary discriminator

H.5 Phase 4 — Consensus Decision

PHASE 4: CONSENSUS | NDI + 3-engine zone vote -> single clinical decision

Condition	Zone	Clinical action
All 3 engines: LOW	LIKELY HEALTHY	Routine follow-up. Voice alone not conclusive.
Any engine: MID OR engines disagree	GREY ZONE	Refer to neurologist. Show full NDI + engine detail.
SVM-RBF AND MLP both MID	GREY ZONE - Hard Case	Senior neurologist flag. 3% of population; 67% PD.
All 3 engines: HIGH	CONFIDENT PD	Urgent clinical follow-up. PD specialist referral.
3 repeats disagree significantly	AUTO GREY ZONE	Override to Grey Zone. Inconsistency is a PD signal.

H.6 Phase 5 — Clinical Output

The app displays a colour-coded result card (green/amber/blue) with: NDI score 0-100, zone label, individual engine votes from all 3 attempts, and recommended action. The clinician can expand to see all 10 engine probabilities and 22 acoustic feature values per attempt.

H.7 Phase 6 — Record and Audit

Field	Details
NDI score	0-100 (AUC-weighted, 10 engines, 3 attempts averaged)
Clinical zone	LOW / MID / HIGH + special flags
Engine votes	P(PD) from each of 3 zone engines x 3 attempts = 9 values
All 10 engines	Full NDI detail x 3 attempts = 30 values
Acoustic features	22 features x 3 attempts = 66 values
Audio reference	Hash of each raw recording (not stored externally)
Inconsistency flag	Whether 3 repeats showed significant within-session variance

H.8 Telemonitoring Results (Canonical — GroupKFold-5 by subject)

Engine	Motor R2	Motor MAE	Total R2	Total MAE	Note
Ridge	-0.2164	7.492	-0.1290	9.524	Best (least negative)
Extra Trees	-0.2398	7.191	-0.4041	10.266	
AEGIS Ensemble	-6.8640	21.296	-7.3561	29.019	Amplifies noise

All R2 negative — confirmed finding. UPDRS is patient-specific; cross-subject regression is infeasible. Regression layer pending PPMI longitudinal data.

H.9 AEGIS-Derived Datasets

Dataset	Shape	Contents
AEGIS_P21_Voice_Derived.csv	195 x 40	22 features + 10 engine P(PD) + NDI + grey zone flag + severity tier
AEGIS_P21_Telemonitoring_Derived.csv	5875 x 50	Tele features + 3 engine predictions + slope indicators

DATA GOVERNANCE: Raw source datasets NEVER leave AEGIS secure storage. Only AEGIS-Derived Datasets go external. Violation = DUA breach = programme destroyed.

H.10 The AEGIS Derived Protocol (Final Section)

FORMAL PROTOCOL v1.0 (07 Aug 2026): The citable AEGIS P21 clinical voice protocol derived from experimental results. All future AEGIS P21 recordings must follow this protocol exactly.

H.10.1 Protocol Statement

AEGIS P21 Derived Clinical Voice Protocol — Version 1.0 (07 August 2026)

A three-attempt sustained vowel phonation protocol for Parkinson's Disease screening using acoustic dysphonia analysis. Derived from the MDVP standard (Little et al., 2007) and extended by AEGIS with multi-attempt, multi-engine consensus decision architecture.

H.10.2 Recording Protocol (Standardised)

Parameter	Specification	Rationale
Attempts	3 (one set)	Single attempt is clinically fragile
Duration per attempt	3-6 seconds	Min 3 sec for stable phonation; beyond 6 sec fatigue degrades signal
Inter-attempt rest	2-3 seconds breath-rest	Prevents fatigue carryover between attempts
Vowel sound	Sustained "aaah" at comfortable pitch and loudness	Maximises laryngeal motor demand; eliminates linguistic variability
Recording spec	44.1 kHz, 16-bit, quiet room or noise-cancelling mic	Matches UCI dataset conditions
Target amplitude	70-80 dB SPL	Below 70 dB: poor signal. Above 80 dB: ADC overload distorts jitter/shimmer
Patient posture	Seated, relaxed, 30 sec silence before first attempt	Eliminates posture and nervousness artefact

H.10.3 Feature Extraction Protocol

Parameter	Specification
Features	22 acoustic features per recording (MDVP standard)
Processing	Each of 3 recordings processed independently
No averaging	Features NOT averaged before engine input; each attempt runs engines separately
Engine input	1 x 22 feature vector; StandardScaler applied before classification

H.10.4 Engine Execution Protocol

Layer	Engines	Output	Aggregation
Layer A - NDI	All 10 AEGIS engines	P(PD) per engine per attempt	AUC-weighted mean -> NDI per attempt -> mean across 3 = final NDI (0-100)
Layer B - Zone	MLP + SVM-RBF + Fuzzy SVM-RBF	LOW / MID / HIGH per engine per attempt	Majority zone per engine across 3 attempts -> 3 votes -> consensus rule

H.10.5 Decision Protocol (6 Rules)

Rule	Condition	Output	Clinical action
1	All 3 engine votes: LOW	Likely Healthy	Routine follow-up. Voice alone is not diagnostic.
2	Any engine vote: MID	Grey Zone	Refer to neurologist. Provide NDI score and engine detail.
3	Engine votes disagree	Grey Zone	Refer to neurologist. Disagreement is itself a signal.
4	SVM-RBF AND MLP both MID	Grey Zone - Hard Case	Senior neurologist flag. 67% PD in this subset.
5	3 attempt zones disagree	Auto Grey Zone	Override to Grey Zone. Within-session inconsistency = PD signal.
6	All 3 engine votes: HIGH	Confident PD	Urgent clinical follow-up. PD specialist referral.

H.10.6 Audit Protocol

Every AEGIS P21 assessment generates: patient identifier, date/time, clinician ID, NDI score (0-100), zone, all 10 engine P(PD) values for each of 3 attempts, 22 acoustic features per attempt, inconsistency flag, audio reference hash. Raw audio stored in AEGIS secure storage only.

H.10.7 Mandatory Clinical Caveats

MANDATORY: AEGIS P21 is a SCREENING tool, not diagnostic. "Likely Healthy" does NOT exclude Parkinson's Disease. "Confident PD" is not a diagnosis. All results must be interpreted by a qualified neurologist with full clinical assessment including UPDRS, motor examination, and imaging.

Known limitations v1.0: (1) 195 recordings from 32 subjects - small sample. (2) All subjects male, ages 46-85. (3) AUC 0.7673 - acceptable for screening only. (4) UPDRS regression pending PPMI data. (5) Multi-modal fusion (drawing + gait) pending.

H.10.8 Protocol Version Control

Version	Date	Changes
v1.0	07 Aug 2026	3-attempt standard. 3-engine zone (MLP + SVM-RBF + Fuzzy SVM-RBF). NDI from 10 engines. Thresholds: LOW below 0.35, HIGH above 0.65.
v4.2 (pending)	TBD	FuzzySVM v4.2 fix COMPLETE (07 Aug 2026). Fuzzy Nu-SVC v4.2 fix COMPLETE — excluded (AUC=0.5632). Drawing Board ResNet-50 pending (Owner machine). Multi-modal NDI pending.

This protocol is self-consistent with canonical run results in this appendix and Section 4.6 of this book.

Appendix I: AEGIS P21 Dataset Certification — Distilled Validation Checklist

DATASET CERTIFICATION DATE: 07 August 2026 | Validator: AEGIS v4.2 | RESULT: 21/21 CHECKS PASSED — CERTIFIED FOR EXTERNAL OFFER UNDER DUA

This appendix documents the mandatory pre-release validation run that must pass before AEGIS P21 derived datasets are offered to any partner institution. An institute receiving the derived dataset can train their own models, learn from the acoustic features, validate AEGIS engine predictions, and build on the NDI scoring. Without this certification, only trained model weights can be shared — and institutions generally will not accept model weights alone, as they cannot learn anything from a black-box model without the data.

WHY THE DATASET MATTERS: A partner institution receiving AEGIS_P21_Voice_Derived.csv gets 195 recordings x 22 acoustic features + 11 engine probabilities + NDI score. They can validate our findings, extend our work, and cite AEGIS in their publications. Trained model weights alone = a black box = no scientific value to them.

I.1 Check A — Dataset File Integrity

Check ID	Description	Expected	Result	Status
A1	Row count	195	195	✓ PASS
A2	Engine probability columns	≥ 10	11	✓ PASS
A3	NDI column present (AEGIS_NDI_0to100)	Present	Present	✓ PASS
A4	DSI column absent (retired in v4.2)	Absent	Absent	✓ PASS
A5	Null values in engine prob columns	0	0	✓ PASS
A6	All P(PD) values in valid range [0,1]	All valid	All valid	✓ PASS

I.2 Check B — Canonical AUC Re-verification (GroupKFold-5 by Subject)

Each engine is re-run from scratch on the raw dataset using the canonical GroupKFold-5 protocol (no subject in both train and test). AUC must match the canonical value within ± 0.01 tolerance. Zero diff = perfectly deterministic (SEED=42 + same GroupKFold splits).

Engine	Canonical AUC	Re-run AUC	Diff	Status
MLP Neural Net	0.7673	0.7673	± 0.0000	✓ PASS
Random Forest	0.7640	0.7640	± 0.0000	✓ PASS
Extra Trees	0.7596	0.7596	± 0.0000	✓ PASS
CatBoost	0.7526	0.7526	± 0.0000	✓ PASS
Gaussian NB	0.7476	0.7476	± 0.0000	✓ PASS
Logistic Regression	0.7202	0.7202	± 0.0000	✓ PASS
XGBoost	0.7164	0.7164	± 0.0000	✓ PASS
Hist Gradient Boosting	0.6909	0.6909	± 0.0000	✓ PASS

Engine	Canonical AUC	Re-run AUC	Diff	Status
KNN	0.6840	0.6840	±0.0000	✓ PASS
SVM-RBF	0.6562	0.6562	±0.0000	✓ PASS
Fuzzy SVM-RBF (v4.2)	0.6443	0.6443	±0.0000	✓ PASS

All 11 engines reproduced exactly. Zero diff confirms: (1) no data leakage, (2) deterministic GroupKFold splits, (3) SEED=42 fixed correctly across all estimators.

I.3 Check C — NDI Column Quality

Check ID	Description	Expected	Result	Status
C1	NDI minimum value	≥ 0	0.0	✓ PASS
C2	NDI maximum value	≤ 100	100.0	✓ PASS
C3	NDI null count	0	0	✓ PASS
C4	Correlation NDI vs status (PD=1)	> 0.30	0.3771	✓ PASS

NDI corr=0.3771 with status confirms it is a valid severity signal — higher NDI corresponds to PD voice profiles. Range 0–100 normalised correctly.

I.4 Final Certification

Category	Checks	Passed	Failed	Verdict
A — File Integrity	6	6	0	✓ CERTIFIED
B — AUC Re-verification	11	11	0	✓ CERTIFIED
C — NDI Quality	4	4	0	✓ CERTIFIED
TOTAL	21	21	0	✓✓✓ ALL PASS

CERTIFICATION STATEMENT (07 Aug 2026): AEGIS_P21_Voice_Derived.csv (195 x 41) has passed all 21 mandatory pre-release checks. This dataset may be offered to partner institutions under a signed Data Use Agreement (DUA). The AEGIS-derived columns (11 engine probabilities, Composite PD Score, NDI, Grey Zone Flag) are the intellectual property of AEGIS Research / Dr Loh Kah Meng. Redistribution without written permission is prohibited.

SCOPE LIMITATION: Certification covers the voice dataset only. Telemonitoring dataset (5875 x 50) has not been independently certified yet — all regression engines show negative R2 (cross-subject barrier confirmed). Pending PPMI longitudinal data for proper regression certification.

I.5 Distilled Retest — AEGIS Derived Features Only

Objective: Confirm that AEGIS_P21_Voice_Derived.csv retains sufficient discriminative signal using only the 11 AEGIS_*_prob columns (no raw UCI acoustic features). Method: GroupKFold-5 by subject, cross_val_predict AUC. Pass threshold: distilled AUC within ±0.05 of canonical (raw-feature AUC). Date: 07 August 2026.

Engine	Canon AUC	Distilled AUC	Delta	Status
MLP Neural Net	0.7673	0.7443	−0.023	✓ PASS
Random Forest	0.7640	0.7163	−0.048	✓ PASS
Extra Trees	0.7596	0.7460	−0.014	✓ PASS
CatBoost	0.7526	0.7371	−0.016	✓ PASS
Gaussian NB	0.7476	0.7149	−0.033	✓ PASS

Logistic Regression	0.7202	0.6729	-0.047	✓ PASS (C=0.01)
XGBoost	0.7164	0.6877	-0.029	✓ PASS
Hist GB	0.6909	0.7069	+0.016	✓ PASS
KNN	0.6840	0.7300	+0.046	✓ PASS
SVM-RBF	0.6562	0.6679	+0.012	✓ PASS
Fuzzy SVM-RBF	0.6443	0.5813	-0.063	✗ MARGINAL

VERDICT: 10/11 engines PASS (91%). MLP (primary) and SVM-RBF (anchor) both pass. FuzzySVM-RBF marginal failure (Δ -0.063) is geometry-driven: fuzzy centroid distances collapse in the compressed probability feature space; underlying SVM-RBF passes (+0.012) confirming signal is retained. AEGIS_P21_Voice_Derived.csv is CLEARED for external offer. DUA note: use $C \leq 0.05$ for linear models on prob features; FuzzySVM best applied to raw acoustic features.

Appendix J: P21 Complete Dataset Registry

The following table summarises all datasets planned, acquired, and pending for AEGIS P21 (Parkinson Augmenting System). Certified datasets may be offered externally under DUA. Pending datasets require Owner action before acquisition can proceed.

Dataset	Source	Size	Type	Status	Notes
UCI Parkinson Voice (Primary)	UCI ML Repository / Little et al. 2007	195 recordings 31 subjects	Voice — Acoustic 22 features	✓ ACQUIRED ✓ CERTIFIED	AEGIS_P21_Voice_Derived.csv (195 × 41) cleared for external offer under DUA
UCI Parkinson Telemonitoring	UCI ML Repository / Tsanas et al. 2010	5,875 recordings 42 subjects	Voice — Telemon. 26 features	✓ ACQUIRED ⚠ PENDING	Cross-subject R^2 all neg. Needs PPMI longitudinal for regression cert.
PPMI (Primary)	Parkinson Progression Markers Initiative ppmi-info.org	TBD (large cohort)	Multi-modal Neuroimaging + biomarker	⌚ PENDING Owner registration	Dean's letter ready. Owner must register at ppmi-info.org
PhysioNet Gait — NDD	PhysioNet / Hausdorff et al.	Multiple subjects gait cycles	Gait — stride time series	⌚ PENDING Owner download	Dean's letter ready. Gait features for P21 expansion.
PhysioNet Gait — PD	PhysioNet / Frenkel-Toledo et al.	PD patients + HC gait cycles	Gait — stride time series	⌚ PENDING Owner download	Paired with Gait-NDD. GroupKFold-5 by subject.
UCI Replicated Acoustic	UCI ML Repository (Sakar et al. 2019)	1,040 recordings 252 subjects	Voice — Acoustic extended (754 features)	⌚ PENDING Institutional email	Requires institutional email for download. Massive feature set.
HandPD	HandPD Dataset (Pereira et al.)	Handwriting images PD + HC	Handwriting spiral/meander	⌚ PENDING Institutional email	Institutional email req. ResNet-50 feature ext.
Drawing Board (Internal)	AEGIS Collected (Controlled acquisition)	408 images spiral + wave	Handwriting spiral/wave	⌚ PENDING Owner GPU run	ResNet-50 on Owner machine required. All images secured.

Acquired: 2 datasets (Voice + Telemonitoring). Certified for external offer: 1 (Voice). Pending Owner/institutional action: 6 datasets. As datasets are acquired, each must complete the 21-check certification protocol (Appendix I) before any external offer is made.

Appendix K: Partner Institution Learning Framework

This appendix defines the recommended learning framework for partner institutions receiving AEGIS_P21_Voice_Derived.csv under the Data Use Agreement (AEGIS-P21-DUA-001). It is provided for review by supervising academic staff. All terms are governed by the DUA.

K.1 What the Partner Can Build

Using the 195-recording derived dataset, a partner institution can build a voice-based Parkinson's disease classifier achieving AUC 0.74–0.77 under honest leave-one-subject-out validation (GroupKFold-5). This is a real, reproducible, leakage-free research benchmark. The AEGIS clinical-grade system (AUC 0.7673 with 3-engine grey zone) is NOT provided; partners receive a learning-grade pathway that teaches the science without exposing AEGIS IP.

K.2 Three-Level Learning Curriculum

Level	Classifier	What it teaches / Expected AUC
1 — Introductory	Decision Tree	Feature splits, interpretability, overfitting. AUC ~0.62–0.65. 5–10 lines of code. Tree can be visualised and printed.
2 — Intermediate	Random Forest + SVM-RBF (comparison)	Ensemble learning, bagging, feature importance. RF AUC ~0.74–0.76. SVM-RBF AUC ~0.65–0.67 — teaches why kernels do not always win. Feature importance ranking shows which acoustic features matter.
3 — Advanced	MLP Neural Network	Non-linear classification, hyperparameter tuning, probability calibration. AUC ~0.74–0.77. Connects conceptually to deep learning and AEGIS architecture. Best result achievable on this dataset.

K.3 Mandatory Rules (Non-Negotiable)

The following rules apply to all experiments. Violation of R1 or R4 constitutes a breach of the DUA.

#	Rule	Why it matters
R1	Use GroupKFold-5 by subject (mandatory)	No subject may appear in both train and test. Random splits give AUC ~0.91 (leakage — inflated). Honest GroupKFold-5 gives AUC ~0.74–0.77. Publishing leaky results discredits the lab.
R2	Handle class imbalance	75% PD, 25% HC. Without class_weight="balanced", the classifier learns to predict all PD and reports 75% accuracy as a false success. Must be addressed in every experiment.
R3	Report AUC, not just accuracy	AUC is the correct metric for imbalanced clinical screening data. Accuracy without sensitivity/specificity breakdown is insufficient for publication.
R4	No Clinical Use — ever	The dataset and any classifier built from it are for learning and research ONLY. Clinical deployment requires separate regulatory validation. See DUA Clause 3.
R5	Cite AEGIS + UCI in all publications	Mandatory attribution: (1) AEGIS Research / Dr Loh Kah Meng — AEGIS-P21-DUA-001. (2) Little et al. 2009, IEEE Trans Biomed Eng. Failure to cite is a breach of the DUA.

K.4 AEGIS IP Boundary

The table below defines what the partner may and may not do. When in doubt, the DUA takes precedence.

Category	What partner CAN do	What partner CANNOT do (AEGIS IP)
Classifiers	Code Decision Tree, RF, SVM, MLP using scikit-learn	Reproduce FuzzyCls (fuzzy membership weighting), 3-engine grey zone architecture, 11-engine board
Scoring	Use AEGIS_NDI_0to100 column as an input feature	Reproduce the NDI formula (AUC-weighted ensemble → 0–100). The formula is not disclosed.
Clinical app	Use AEGIS app as a reference/fallback during research	Claim equivalence to AEGIS app, deploy a competing clinical tool based on this dataset
Publication	Publish research findings with AEGIS + UCI citations	Claim AEGIS methodology as their own, omit attribution, publish with leaky validation

K.5 AEGIS Clinical Fallback

AEGIS operates openly. The AEGIS P21 Smart App is available as a clinical reference instrument during the partner's research and development phase. Partners may use the app to compare their classifier's outputs against AEGIS clinical-grade results on the same voice recordings. We share the data and the tools because science advances when institutions can build, compare, validate, and cite — not when knowledge is hoarded. The DUA protects AEGIS IP while enabling genuine academic collaboration.

AEGIS-P21-DUA-001 · Appendix K · FOR SUPERVISING ACADEMIC REVIEW — NOT FOR CLINICAL USE

Appendix L: Can a Machine Hear Parkinson's Disease?

A Laboratory Exercise Series • AEGIS P21 Voice Classification • Three Levels, One Discovery

Designed for undergraduate and postgraduate students in biomedical engineering, data science, and clinical informatics.

Imagine you could hear Parkinson's Disease. Not see it on a scan. Not read it in a blood result. Hear it — in the tiny tremors woven through a sustained vowel, in the fragile breath control behind a simple "aaah", in patterns so subtle no human ear can catch them, but mathematics can. This is not science fiction. The data is in front of you right now. This laboratory will show you exactly how it works.

The dataset — 195 voices, 22 measurements, one question

Between 2002 and 2004, researchers at the University of Oxford recorded 195 voice samples from 32 individuals. Some were living with Parkinson's Disease. Some were not. Each recording was reduced to 22 acoustic measurements — jitter, shimmer, harmonic-to-noise ratio, and others — numbers that capture the invisible physics of a human voice struggling, or not, to stay steady. That dataset has since been downloaded hundreds of thousands of times. It has been the subject of over a thousand published papers. And now it is in your hands.

147 of those 195 recordings come from people with Parkinson's Disease. 48 come from healthy controls. Your mission: build a classifier that can tell them apart. By the end of these exercises, you will have built three. Each one teaches you something the others cannot.

These exercises are provided exclusively for academic learning under the AEGIS Data Use Agreement (AEGIS-P21-DUA-001). No classifier developed here may be used for clinical diagnosis, patient screening, or any medical application. Real clinical-grade deployment is a separate endeavour requiring separate validation. The AEGIS clinical system is available as a reference — see the closing note.

Before you begin

You will need: Python 3.8+ • scikit-learn • pandas • numpy • AEGIS_P21_Voice_Derived.csv

```
pip install scikit-learn pandas numpy
```

Estimated time: 90–120 minutes total • Work through the labs in order — each one builds on the last.

Lab 1 — The Decision Tree: Teaching a Machine to Think Like a Clinician

Level: Beginner · Time: ~25 minutes · Code: ~15 lines

A clinician at the bedside does not run a regression. They ask questions. "Is the tremor resting or intentional?" "Does the voice break on sustained phonation?" "Is there micrographia?" A decision tree does exactly this — it learns which questions to ask, in which order, to separate PD from healthy. The beauty: you can print the result on a single page and read every decision it makes. There is nothing hidden. Nothing opaque. Just logic, learned from data.

What the machine will learn to ask

Each branch of your tree will be a threshold on one acoustic measurement. At the top — the root — it will ask the single most discriminating question in the entire dataset. Work your way down and you will understand, branch by branch, what separates a Parkinson's voice from a healthy one. No black boxes. No mysteries. Just mathematics you can trace with a finger.

The code skeleton — fill every blank marked ____

```
import pandas as pd
from sklearn.tree import DecisionTreeClassifier, export_text
from sklearn.model_selection import ____, cross_val_predict # GroupKFold
from sklearn.metrics import roc_auc_score

# Load the dataset
df = pd.read_csv("AEGIS_P21_Voice_Derived.csv")
feat = [c for c in df.columns if c.startswith("AEGIS_") and c.endswith("_prob")]
X = df[feat].values
y = df["status"].values # 1 = Parkinson's, 0 = Healthy
grps = df["name"].str.rsplit("_",n=1).str[0].values # one ID per subject

# CRITICAL: honest validation — no subject appears in both train and test
gkf = ____ (n_splits=5) # GroupKFold

# Build the tree — try depth 3, 4, 5 and compare
model = DecisionTreeClassifier(max_depth=____, class_weight="balanced",
random_state=42)

# Run cross-validation
proba = cross_val_predict(model, X, y, groups=grps, cv=gkf, method="predict_proba")
auc = roc_auc_score(y, ____[:,1]) # proba
print(f"AUC: {auc:.4f}")

# Fit on full data — then read the tree
model.fit(X, y)
labels = [f.replace("AEGIS_", "").replace("_prob", "") for f in feat]
print(export_text(model, feature_names=labels, max_depth=4))
```

#	Question	Your answer
Q1	What AUC does your tree achieve at max_depth=4? Is it above 0.65?	AUC = _____
Q2	What is at the ROOT of your tree — the very first question it asks? Why do you think that feature was chosen first?	Root = _____
Q3	What happens to AUC as you increase max_depth from 3 to 6? At what depth does it peak, then decline? What is the name for this phenomenon?	Peak at depth = _____
Q4	Are PD cases or healthy cases harder to classify correctly? Look at the confusion matrix. What might explain the	Harder: _____

	imbalance?	
Q5 ★	WHY do we use GroupKFold instead of random splitting? What inflated AUC would you get with random splits — and why is that scientifically dishonest?	Random split AUC ≈ _

Deep challenge — remove `class_weight="balanced"`. Run again. Watch what happens to the healthy controls. The machine will quietly stop caring about them and report a suspiciously high accuracy. This is one of the most dangerous traps in medical machine learning. You have just sprung it — deliberately, safely, in a lab.

Lab 2 — Random Forest vs SVM: 300 Opinions vs One Perfect Boundary

Level: Intermediate · Time: ~30 minutes · Code: ~30 lines

Your decision tree was one doctor. Occasionally brilliant, occasionally overconfident. A random forest is 300 doctors, each trained on a slightly different view of the same data, voting together. Meanwhile a Support Vector Machine takes a completely different philosophy: it searches for the single widest possible gap between PD and healthy voice profiles — the mathematical equivalent of finding the cleanest possible dividing line. Both approaches have won Nobel-level breakthroughs in their fields. Which one wins here?

Part A — The Random Forest

300 trees. Each sees a random subset of features and a random subset of recordings. Each votes. The majority decides. And as a bonus: after voting, the forest will tell you which features it relied on most heavily. That ranking is one of the most actionable insights in the entire exercise.

```
from sklearn.ensemble import RandomForestClassifier

rf = RandomForestClassifier(
    n_estimators=300,
    class_weight="balanced",
    random_state=42
)
proba_rf = cross_val_predict(rf, X, y, groups=grps, cv=gkf,
                             method="predict_proba")
auc_rf = roc_auc_score(y, proba_rf[:,1])
print(f"Random Forest AUC: {auc_rf:.4f}")

# Which engines does the forest trust most?
rf.fit(X, y)
ranked = sorted(zip(feats, rf.feature_importances_), key=lambda x: -x[1])
for name, imp in ranked:
    bar = chr(9608) * int(imp * 50)
    print(f" {name.replace('AEGIS_', '').replace('_prob', '')}<22} {imp:.3f}
{bar}")
```

Part B — The Support Vector Machine

The SVM does not think about probabilities or votes. It finds the hyperplane that maximises the margin between the two classes. With an RBF kernel it can curve that boundary to fit complex shapes in the data. The question is: does the extra geometric sophistication pay off?

```
from sklearn.svm import SVC
from sklearn.preprocessing import StandardScaler
from sklearn.pipeline import Pipeline

# Always scale before SVM — the kernel is sensitive to feature magnitude
svm = Pipeline([
    ("sc", StandardScaler()),
    ("m", SVC(kernel="rbf", probability=True, class_weight="balanced",
              random_state=42))
])
proba_svm = cross_val_predict(svm, X, y, groups=grps, cv=gkf,
                              method="predict_proba")
auc_svm = roc_auc_score(y, proba_svm[:,1])
print(f"SVM-RBF AUC: {auc_svm:.4f}")
```

Fill in your results

Model	AUC	Beats Decision Tree?	Most trusted feature (RF only)
Decision Tree (Lab 1)	___	Baseline	_____

Random Forest	___	Yes / No	_____
SVM-RBF	___	Yes / No	N/A

#	Question	Your answer
Q1	Does RF beat SVM? By how many AUC points? Does this surprise you, given SVM's mathematical elegance?	Gap = _____
Q2	The feature importance ranking reveals which AEGIS engine the forest trusts most. Is this the same engine that achieved the highest AUC on the full voice board (see Section 3 of the book)?	Match? Yes / No
Q3	Try RF with n_estimators = 10, 50, 100, 200, 300. At what point do additional trees stop improving AUC? Plot the curve.	Plateau at n = _____
Q4 ★	You now have three models at three AUC levels. What is the pattern? Write one sentence that explains why each step up in model complexity does or does not deliver better performance on this particular dataset.	(reflection)

Boardroom question: Your department is asked to justify investment in AI-based PD screening. A trustee asks: "Why not just use SVM — it's simpler?" Using your results from this lab, write a two-sentence answer that is technically honest and non-dismissive of the question.

Lab 3 — The Neural Network: Certainty, Doubt, and the Grey Zone

Level: Advanced · Time: ~45 minutes · Code: ~40 lines

Here is what none of your previous models could tell you: "I am not sure." A neural network outputs a probability — not a label. $P(\text{PD}) = 0.97$ means confidence. $P(\text{PD}) = 0.52$ means genuine doubt. And in a clinical context, genuine doubt is not a failure. It is the machine's way of saying: "This patient needs a human expert, not an algorithm." That insight — knowing the limits of your own certainty — is the difference between a toy and a tool.

Part A — Build and evaluate the MLP

```
from sklearn.neural_network import MLPClassifier
from sklearn.metrics import confusion_matrix, roc_curve, roc_auc_score
import numpy as np

mlp = Pipeline([
    ("sc", StandardScaler()),
    ("m", MLPClassifier(
        hidden_layer_sizes=(64, 32), # two hidden layers
        max_iter=500,
        early_stopping=True,          # stops when it stops learning
        random_state=42
    ))
])

proba_mlp = cross_val_predict(mlp, X, y, groups=grps, cv=gkf,
                              method="predict_proba")
auc_mlp = roc_auc_score(y, proba_mlp[:,1])
print(f"MLP AUC: {auc_mlp:.4f}")
```

Part B — Sensitivity, Specificity, and the clinical tradeoff

In a screening context, two errors carry different weights. Missing a PD patient means delayed diagnosis and delayed treatment. A false positive means unnecessary anxiety and follow-up. You cannot minimise both simultaneously — but you can choose where on that curve you stand.

```
# Standard threshold: 0.5
preds_05 = (proba_mlp[:,1] >= 0.5).astype(int)
tn, fp, fn, tp = confusion_matrix(y, preds_05).ravel()
sensitivity = tp / (tp + fn) # PD cases correctly caught
specificity = tn / (tn + fp) # healthy controls correctly cleared
print(f"Sensitivity: {sensitivity:.4f} Specificity: {specificity:.4f}")

# Find the threshold that maximises (sensitivity + specificity)
fpr, tpr, thresholds = roc_curve(y, proba_mlp[:,1])
best_idx = np.argmax(tpr - fpr)
best_thr = thresholds[best_idx]
print(f"Optimal threshold: {best_thr:.3f}")

# Apply optimal threshold and recompute
preds_opt = (proba_mlp[:,1] >= best_thr).astype(int)
tn2, fp2, fn2, tp2 = confusion_matrix(y, preds_opt).ravel()
print(f"Optimal sensitivity: {tp2/(tp2+fn2):.4f} specificity: {tn2/(tn2+fp2):.4f}")
```

Part C — The Grey Zone

The recordings where the machine hesitates are the most clinically interesting. They are not necessarily the machine's failures — they may be the cases where even a specialist would pause. Find them. Study them.

```
# Recordings where  $P(\text{PD})$  falls in the uncertain middle
```

```
grey = (proba_mlp[:,1] >= 0.35) & (proba_mlp[:,1] <= 0.65)
grey_df = df[grey].copy()
grey_df["P_PD"] = proba_mlp[grey, 1]
```

```
print(f"Grey zone: {grey.sum()} recordings ({grey.mean():.1%} of total)")
print(f"Of those: {grey_df['status'].sum()} are actually PD  
({grey_df['status'].mean():.1%}")
print(f"Grey zone sorted by uncertainty:")
print(grey_df[["name", "status", "P_PD"]].sort_values("P_PD").to_string())
```

#	Question	Your answer
Q1	MLP AUC vs your previous best model. Is the improvement large or modest? What does that tell you about diminishing returns in model complexity?	MLP AUC = _____
Q2	At threshold 0.5: sensitivity and specificity. Which is higher? In a screening programme where missing PD is the greater harm, which would you prioritise?	Sen=____ Spec=____
Q3	At the optimal threshold: does sensitivity improve? What happens to specificity? Is the tradeoff clinically acceptable? Justify your answer.	_____
Q4	How many recordings fall in the grey zone? What fraction are actually PD? Does the grey zone behave like a random sample of the dataset, or is it skewed?	n=____ PD%=_____
Q5 ★ ★	The machine flags grey zone cases as uncertain. A clinician colleague says: "The AI doesn't know — so it's useless for these patients." Write a counter-argument. Is uncertainty a sign of failure, or a sign of clinical wisdom?	(open reflection)

Grand challenge: Can you reduce the grey zone without sacrificing AUC? Try adjusting hidden_layer_sizes — e.g. (128,64,32), (32,), (256,128). Plot grey zone size vs AUC. Spoiler: you will find they are in tension. The machine that claims to be most certain is not always the most accurate. This is the central philosophical problem of medical AI.